
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is a widely used neighbor embedding method whose visualizations are produced by attractive and repulsive forces. This paper makes those forces explicit by introducing attraction and repulsion shapes: scalar functions that reveal how each update contracts or expands distances in the embedding. The analysis yields four reviewer-facing contributions. First, we derive contraction and expansion conditions that explain why UMAP's attraction can simultaneously act as attraction and repulsion at short distances. Second, we use these shapes to explain the role of learning-rate annealing and to compare UMAP with NEG- t -SNE, Parametric UMAP, PaCMAP, and related methods. Third, we modify attraction to improve consistency under random initialization, showing that stronger long-range attraction helps recover a stable structure. Fourth, we isolate the role of repulsion in controlling compactness and inter-cluster distance. Together, these results make existing UMAP-style embeddings more interpretable and provide shape-level tools for improving robustness and visualization quality.

1 Introduction

High-dimensional data are now routine, but their structure is difficult to inspect directly. Dimensionality reduction (DR) methods address this problem by producing low-dimensional views for exploratory analysis. Two of the most widely used methods, t -distributed stochastic neighbor embedding (Maaten and Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018), rely on an attraction–repulsion principle: similar points are pulled together, while dissimilar points are pushed apart. Because these algorithms are unsupervised, their value depends on whether the resulting geometry can be interpreted without labels. Their attractive and repulsive relationships make them useful for clustering (Becht et al., 2019), exploratory data analysis (Fleischer and Islam, 2020), anomaly detection in semiconductor manufacturing (Fan et al., 2021), visual search (González-Márquez et al., 2024), time series analysis (Altın and Cakir, 2024), studying representation convergence (Huh et al., 2024), and outlier image detection (Islam and Fleischer, 2024). The central question for reviewers and practitioners is therefore not only whether these methods produce clusters, but which force-level mechanisms make those clusters appear, move, or disappear.

Newer neighbor-embedding methods extend this same principle in different directions (Amid and Warmuth, 2019; Agrawal et al., 2021; Wang et al., 2021; Narayan et al., 2021; Yang et al., 2022; Wang et al., 2025; Kury et al., 2025). Yet the practices that make them work—initialization choices, learning-rate schedules, and kernel-parameter tuning—often remain heuristic, even though DR necessarily introduces distortions (Chari and Pachter, 2023). This paper targets that interpretability gap. Rather than treating compact clusters as a black-box consequence of optimization, we ask which part of the force law controls learning-rate sensitivity, which part controls consistency under random initialization, and which part controls cluster compactness and separation.

Our contribution is a shape-level analysis of attraction and repulsion. We decompose each update into a distance-dependent scalar shape and a displacement vector, then use the shape to identify when an update contracts or expands distances. This makes the paper's contributions explicit: attraction explains learning-rate annealing, random-initialization sensitivity, and much of cluster formation, while repulsion primarily regulates compactness and inter-cluster distance. Specifically, we contribute:

1. We formulate attraction and repulsion shapes from the attractive and repulsive forces, establish the conditions for contraction and expansion of distance, and provide a fresh perspective for these algorithms (Section 4).

2. We show that the attraction shape of UMAP causes the counterintuitive concept of both contraction and expansion of distance. Comparing attraction shapes of different algorithms, we discuss how attraction influences the learning rate annealing scheme (Section 5).
3. We modify the attraction shape to improve the consistency of embedding under random initialization. This indicates the encoding of a unique structure (Section 5.1).
4. Analyzing repulsion shapes, we provide a deeper understanding of cluster formation and regulating inter-cluster distance (Section 5.2).
5. We compare attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG-*t*-SNE, unveil the similarities and distinctions among them, and characterize the stability of the algorithms (Section 6).

The main text therefore uses UMAP as the primary case study and brings in other algorithms when they sharpen the contribution being made. The appendix extends the same shape analysis to additional methods, datasets, metrics, and implementation details. The codes used in the research will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor embedding begins with stochastic neighbor embedding (SNE) (Hinton and Roweis 2002) and its *t*-distributed extension (*t*-SNE) (Maaten and Hinton 2008). These methods use dense, normalized pairwise graphs, so every point interacts with every other point through probability-like weights. In parallel, locally linear embedding (Roweis and Saul 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi 2002) used *k*-nearest neighbor (*k*-NN) graphs and spectral decompositions. Tang et al. (Tang et al. 2016) brought the *k*-NN graph into an iterative optimization and removed low-dimensional normalization; UMAP (McInnes et al. 2018) then removed normalization in both spaces and optimized pairwise interactions directly. This history motivates our contribution: the algorithms differ in graph construction and normalization, but their optimization can still be read through explicit attractive and repulsive forces. PaCMAP (Wang et al. 2021), NEG-*t*-SNE (Damrich et al. 2023), and related methods refine this force design; a recent survey appears in de Bodt et al. (2025).

Prior theory has clarified when and why these methods preserve clusters. For SNE, well-clustered data can remain well-clustered after embedding (Shaham and Steinerberger 2017); for *t*-SNE, the number of embedded clusters can lower-bound the number of original clusters (Linderman and Steinerberger 2019), with additional characterizations in (Arora et al. 2018; Cai and Ma 2022; Linderman and Steinerberger 2022). Another line of work connects methods that appear visually different (Böhm et al. 2022; Damrich et al. 2023): LE, ForceAtlas2 (Jacomy et al. 2014), UMAP, and *t*-SNE can be viewed as points on a common spectrum (Böhm et al. 2022); LE and *t*-SNE are linked through early exaggeration (Cai and Ma 2022); and UMAP and *t*-SNE are related through contrastive estimation (Damrich et al. 2023). Related work also connects SNE to contrastive learning (Hu et al. 2023), probabilistic formulations (Ravuri et al. 2023; Ravuri and Lawrence 2024), kernel views (Draganov et al. 2023; Draganov and Dohn 2023), and information geometry (Kolpakov and Rocke 2024). Our work complements these explanations by making the local force shapes themselves the unit of analysis.

3 Uniform Manifold Approximation and Projection

We begin with UMAP because it exposes the force mechanisms that the paper analyzes. UMAP constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ through a pairwise relation $p_{i,j} = f_h(d(x_i, x_j))$, typically with values in $[0, 1]$ (for details, see Appendix H).

The low-dimensional graph for $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is defined by the differentiable affinity

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d regulates the distance between the two nearest low-dimensional points. This parameterization matters for our contribution because the force shapes studied below are induced by this affinity.

UMAP minimizes the cross-entropy loss

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term is the source of attraction and the second term is the source of repulsion. This separation is the methodological entry point of the paper: once the two terms are written as forces, their scalar shapes can be analyzed independently. UMAP does not optimize all pairs at every iteration; instead, it uses negative sampling (Mikolov et al., 2013, Tang et al., 2016). For each edge with $p_{ij} > 0$, a positive edge, the algorithm samples several negative edges at random. The attractive update acts on the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

whereas the repulsive update acts on the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number. The point y_j is not updated for negative edges. This standard formulation sets up the rest of the paper: we keep the scientific model fixed and reinterpret its updates through attraction and repulsion shapes. For a detailed analysis of UMAP’s loss, see (Damrich and Hamprecht, 2021).

4 Attraction and Repulsion Shapes

The first contribution is a force-shape formalism. We rewrite the updates (4)-(6) by decomposing each gradient, $\nabla_{y_i^t} \log(q_{ij})$ or $\nabla_{y_i^t} \log(1 - q_{ij})$, into a distance-dependent scalar coefficient and the displacement vector $(y_i - y_j)$. We call the scalar coefficient the attraction shape or repulsion shape. Although UMAP is the running example, the same decomposition applies to any neighbor embedding whose optimization is organized around attraction and repulsion.

By writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, we can update the equations of a positive edge as

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and the Euclidean metric gives $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. The same shape-level view rewrites a negative edge as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

This decomposition has appeared in prior work (McInnes et al., 2018, Agrawal et al., 2021, Draganov and Dohn, 2023), but the contribution here is to make the scalar functions the object of study. The original UMAP paper (McInnes et al., 2018) used the derivative form for efficient computation, and (Draganov and Dohn, 2023) used related derivatives to compare kernels. Similarly, (Agrawal et al., 2021) decomposed forces into a scalar coefficient and a unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ to emphasize magnitude. We instead retain the displacement vector and analyze the scalar coefficient as a shape, because the shape directly reveals when a nominally attractive or repulsive update changes distance in the intended direction.

4.1 Conditions for Attraction and Repulsion

We now turn the shape formalism into a reviewer-checkable criterion. From the update equations (7)-(10), the following proposition states exactly when a positive edge contracts the distance between y_i^t and y_j^t :

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Thus λf_a is the effective attraction shape, although f_a alone is often enough to interpret the embedding. For negative edges, the companion condition identifies when the update expands the distance between y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

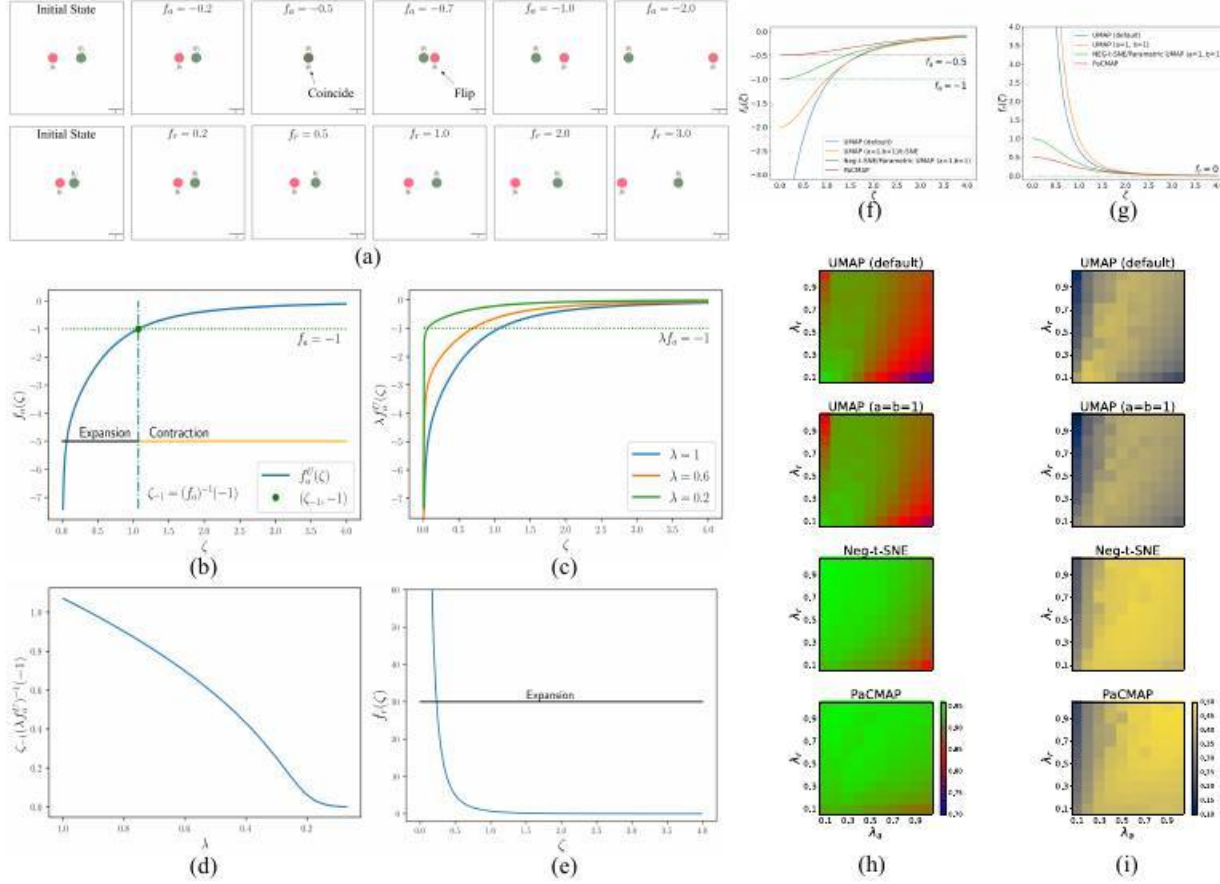


Figure 1: Shape-level diagnostics for UMAP and related embeddings. (a) Values of f_a (top) and f_r (bottom) determine whether a pair contracts or expands. (b) UMAP’s attraction shape reveals the short-distance expansion regime. (c) Effective attraction (λf_a) changes with learning rate λ . (d) The minimum contraction distance (ζ_{-1}) decreases as λ decreases. (e) UMAP’s repulsion shape controls short-range separation. (f) Attraction and (g) repulsion shapes compare the force designs of several embedding methods. (h) Trustworthiness and (i) Silhouette Score show how independently varying attraction (λ_a) and repulsion (λ_r) affects embedding quality. Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

Adding a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ to Eq. (10) would not change this conclusion. These two propositions are therefore the paper’s main diagnostic tool: they give per-iteration certificates for contraction and expansion, separate from any learning-rate tuning or decay mechanism. In principle, one can design shapes whose contraction and expansion behavior does not depend on annealing.

Figure 1(a) illustrates why the shape view is useful. A positive repulsion shape is straightforward: any $f_r > 0$ increases the distance. Attraction is less intuitive. For $f_a \in (-1.0, 0)$, the distance contracts, with the strongest coincident-point update at $f_a = -0.5$; values below -1.0 instead expand the distance. This local distinction is the basis for the later contributions on learning-rate annealing, initialization consistency, and cluster formation.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

The second contribution is an explanation of UMAP through its default shapes. Applying the gradient decomposition to the affinity in (1), UMAP’s attraction and repulsion shapes are

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

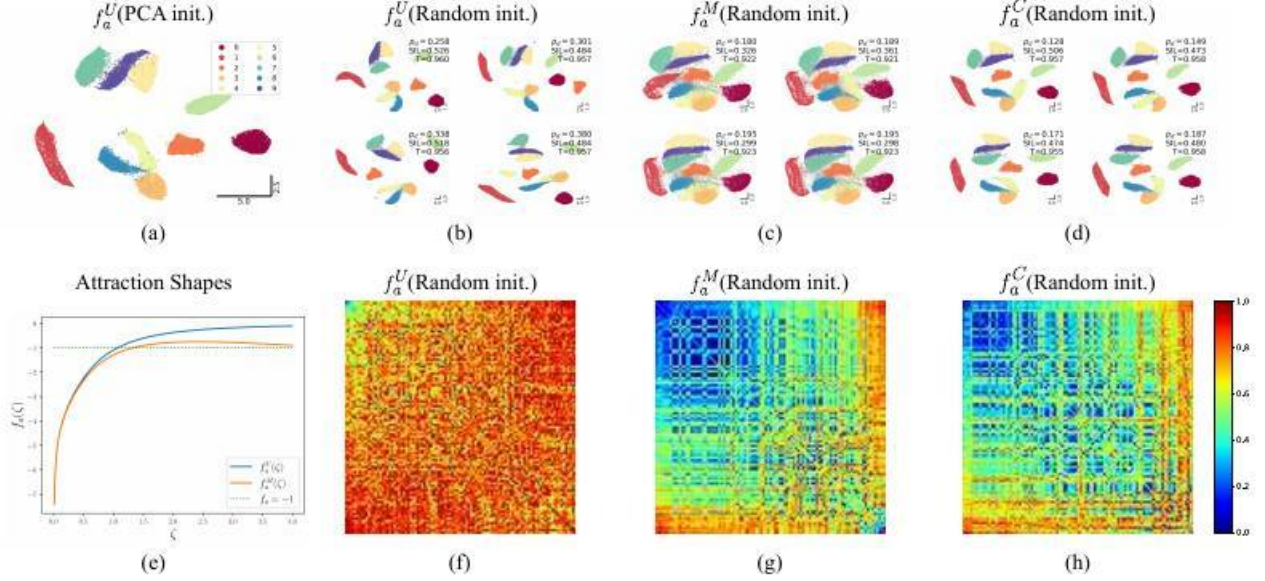


Figure 2: Initialization-consistency contribution on MNIST. (a) PCA-initialized reference mapping. (b-d) Four random-initialized mappings with the lowest Procrustes distance (p_d) to (a) for (b) default UMAP, (c) modified attraction, and (d) composite attraction. (e) Default and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default UMAP (0.78 ± 0.13), (g) modified attraction (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20). Diagonal (i, i) entries are sorted by p_d from (a), while off-diagonal (i, j) entries give p_d between the i^{th} and j^{th} mappings. The matrices and (mean $p_d \pm \text{std}$) values show that default UMAP embeddings are not self-similar, whereas modified and composite attraction shapes encourage scale-invariant structure.

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. We use these functions to make the roles of a and b explicit. The following discussion is generally valid for $b \leq 1$, where the attraction shape is strictly increasing and therefore invertible (for derivation, see Appendix A).

Figure 1(b) shows the default UMAP attraction shape ($a = 1.58, b = 0.89$). Its magnitude diverges as $\zeta \rightarrow 0$, and Proposition 4.1 identifies the transition from contraction to expansion when λf_a^U crosses -1 . Because the attraction shape is invertible, we define $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$ as the minimum distance for contraction under attractive updates. Distances $\zeta > \zeta_{-1}$ contract, whereas $0 < \zeta < \zeta_{-1}$ expand. This is the first central finding: UMAP’s attractive update can repel nearby points.

A large ζ_{-1} makes neighboring points oscillate between contraction and expansion, producing fuzzy clusters. The shape analysis therefore reframes annealing: sharp boundaries require driving $\zeta_{-1} \rightarrow 0$, which UMAP accomplishes by reducing the learning rate toward zero (Figs. 1(c,d)). Repulsion behaves differently. The repulsion shape in Eq. 12 is always positive, satisfies Proposition 4.2, approaches 0 as $\zeta \rightarrow \infty$, and diverges as $\zeta \rightarrow 0$ (Fig. 1(e)).

Comparing attraction shapes across algorithms makes the contribution broader than UMAP. Only UMAP and t-SNE face the problem $f_a < -1$ (and therefore $\zeta_1 > 0$ at $\lambda = 1$, Fig. 1(f)). t-SNE mitigates this through the pair weights p_{ij} , whereas UMAP relies on learning-rate annealing. By contrast, Neg-t-SNE and PaCMAP keep f_a naturally in $[-1, 0]$, so Proposition 4.1 holds for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$; PaCMAP’s weighted f_a is always below 0.5, preventing attraction-induced flips. The repulsion comparison shows that UMAP is distinctive in having very large repulsion at small distances (Fig. 1(g)); in implementations, these unbounded values are often clipped. Appendix G extends this comparison to additional methods.

We test this interpretation on MNIST by assigning separate constant learning rates to attraction (λ_a) and repulsion (λ_r) and varying them independently (Figs. 1(h,i)). For UMAP, smaller attraction learning rates (< 0.5 , making $\zeta_{-1} \simeq 0$) improve both trustworthiness (T) (Venna and Kaski 2001), which measures structure preservation, and silhouette score (SIL) (Rousseeuw 1987), which measures label separation. In contrast, the full range $\lambda_r \in [0, 1]$ remains usable. For NEG-t-SNE and PaCMAP, where $\zeta_{-1} = 0$ by shape design, the full range of tested parameters is effective. This result supports the contribution claim: UMAP’s embedding and clustering quality depend on reducing ζ_{-1} , usually

through annealing, whereas newer algorithms build this property into their attraction shapes. Additional datasets and illustrations appear in Section 6 and Appendices B, G and I.

5.1 Improving UMAP’s Consistency under Random Initialization

The third contribution uses the shape formalism to address consistency under random initialization. UMAP is known to depend on good initialization (Kobak and Linderman 2021, Wang et al. 2021): PCA initialization or spectral initialization can produce consistent mappings despite stochastic optimization, whereas random Gaussian initialization can cause clusters to fail or to appear in different orientations across runs. The shape explanation is near-sightedness (Wang et al. 2021). If two high-dimensional neighbors start far apart in the embedding, then $|f_a^U|$ is too small to pull them together because $|f_a^U|$ diminishes as distance increases (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)|\zeta = 0$).

The shape-level intervention is to induce “far-sightedness”: we increase attraction at large distances so faraway high-dimensional neighbors can still approach each other. To test this hypothesis, we modify UMAP’s attraction shape as

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β controls the strength of the added term (we used $\beta = 0.2$, Fig. 2(e)). In loss-function terms, this shape change adds a regularizer to the attractive term, $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$. Functionally, the modification both pulls distant neighbors together and encourages point intermixing that helps convergence from random initialization, analogous to early exaggeration in *t*-SNE. Equation 13 uses the simplest linear correction; alternatives such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$) may also be suitable.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape uses the modified attraction only early in optimization and then returns to the original UMAP shape, aiming to retain the consistency benefits while reducing distortions from f_a^M . We next evaluate whether these modified and composite shapes make random-initialized embeddings more reproducible.

We evaluate this contribution by first computing a PCA-initialized MNIST embedding (Fig. 2(a)) and then running 100 Gaussian-random initializations for each attraction shape. Procrustes analysis (Gower, 1975) aligns point clouds under scaling, translation, rotation, and reflection (for details, see Appendix D.1); we align each random-initialized embedding to the PCA-initialized reference and report the Procrustes distance (p_d). Figure 2(b) shows the four default-UMAP runs with the lowest p_d : their cluster shapes are similar, but their placements differ. The modified and composite attraction shapes yield visibly more consistent cluster placements (Figs. 2(c,d)).

For a run-to-run summary, we use a Procrustes matrix: diagonal entries are sorted by p_d to the PCA-initialized mapping, and off-diagonal entries are p_d between two random-initialized mappings (for details, see Appendix D.1). This is analogous to a similarity matrix (Foote, 1999). Default UMAP embeddings are not mutually similar (Fig. 2(f)), whereas the modified and composite shapes produce stronger self-similarity (Figs. 2(g,h)). The result supports the paper’s third contribution: UMAP appears to encode a stable structure, but weak long-range attraction can prevent random initializations from reaching it.

5.2 Cluster Formation and Compactness

The fourth contribution separates cluster formation from compactness. In standard UMAP, the most visible user control is the minimum distance parameter m_d in Eq. 2. However, varying m_d restricts the explored values of a and b (Fig. 3(a)), and reducing m_d often yields little additional compactness (Figs. 3(b-d)). The shape explanation is that m_d only weakly changes the repulsion shape (Fig. 3(e)). Directly varying a and b exposes a broader range of behavior. Decreasing a increases repulsion but weakens attraction even faster, worsening near-sightedness; decreasing b provides better control over both shapes (Figs. 3(e,i)). Setting b to 0.6, 0.5, and 0.4 increases inter-cluster distance and breaks previous clusters apart (Figs. 3(f-h)), consistent with heavier-tailed kernels producing smaller, distinct clusters (Van der Maaten and Hinton 2008; Kobak et al. 2019); see also Kobak et al. (2019). Because changing b alters attraction and repulsion simultaneously, this experiment motivates the next controlled separation.

To isolate repulsion, we keep attraction fixed and add a small positive offset (ε) to the repulsion shape:

$$f_r^M = f_r^U + \varepsilon, \quad (15)$$

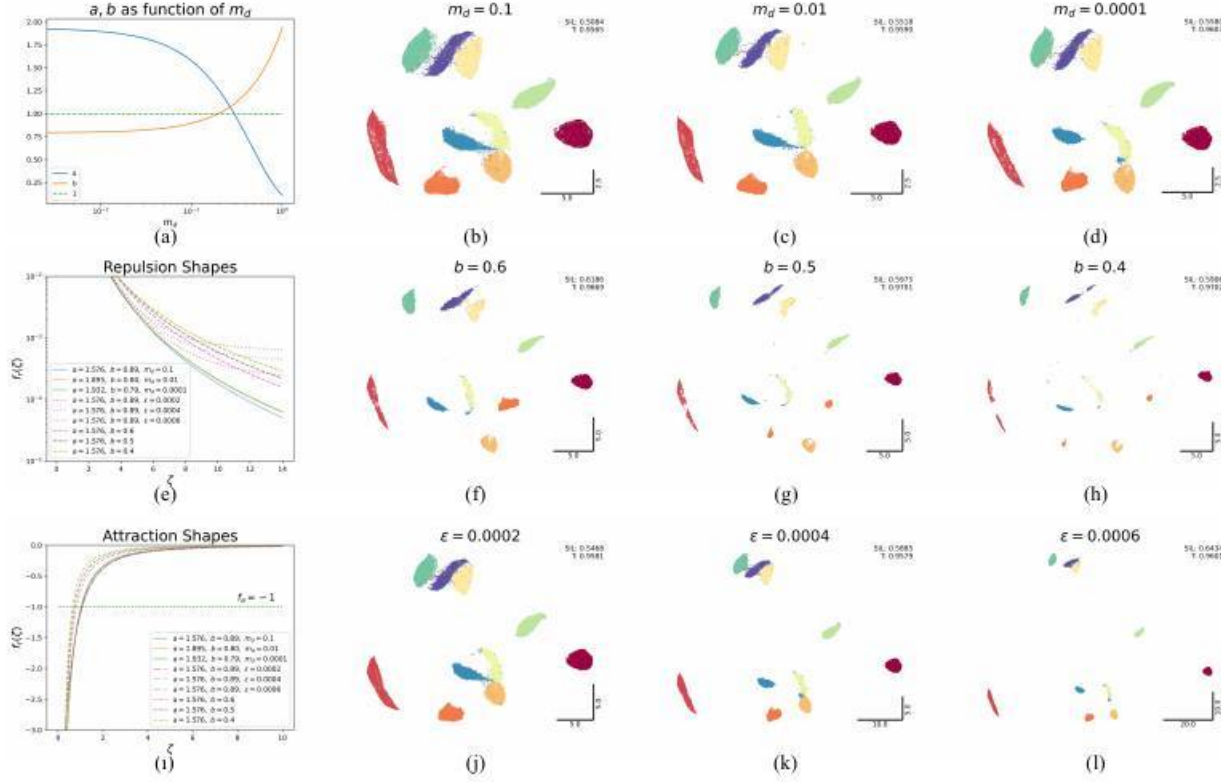


Figure 3: Separating compactness from cluster formation on MNIST. (a) Computing a, b through the low-dimensional distance m_d restricts the explored shape family. (b-d) Setting m_d to 0.1, 0.01, and 0.0001 yields little additional cluster compactness. (e) Repulsion shapes under different parameter choices. (f-h) Explicitly varying b increases repulsion, makes clusters more compact, and reveals clusters absent otherwise. (i) Attraction shapes under the same parameter changes. (j-l) Adding a small positive offset (ϵ) to the repulsion shape increases inter-cluster distance while leaving attraction fixed.

with a and b held fixed. This is equivalent to adding a regularizer to the repulsive term of the loss, $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \epsilon/2 \|y_i - y_j\|_2^2$, and it produces stronger repulsion without changing attraction (Fig. 3(e)). As ϵ increases, inter-cluster distances increase (Figs. 3(j-l)). However, the clustering pattern resembles what is obtained by varying m_d , and larger ϵ yields only loose separation of all labels. This controlled result supports the contribution claim: repulsion regulates distance and compactness, whereas cluster formation is governed predominantly by attraction.

We further disentangle the mechanisms by changing only one shape at a time while leaving the other at the default, using the $b = 0.4$ shape as the intervention (Fig. 4). When attraction remains default and only repulsion is increased, the embedding preserves the default UMAP clusters but makes them smaller and more compact (Fig. 4(a)). When repulsion remains default and attraction is increased, additional within-cluster structure appears and some existing clusters split (Fig. 4(b)). This experiment makes the contribution explicit: attraction drives cluster formation, while repulsion controls compactness and inter-cluster distance. Appendix G.3 shows that LocalMAP exploits the same interplay.

6 Comparison to NEG- t -SNE

The final main-text contribution compares UMAP with NEG- t -SNE and Parametric UMAP through the same shape lens. Many dimensionality reduction algorithms set $a = 1$ and $b = 1$, a choice that often bounds gradients and simplifies optimization. Building on this setting, Damrich et al. (2023) proposed Neg- t -SNE as a stable-gradient alternative to UMAP that can produce compact clusters even at constant learning rate 1. Parametric UMAP (Sainburg et al. 2021) began with the original UMAP loss, but later it adopted a numerically stable modified cross-entropy loss with a logsigmoid kernel (Shi et al. 2023). Shape analysis shows that these two paths lead to the same formulation:

github.com, merged on April 26, 2022

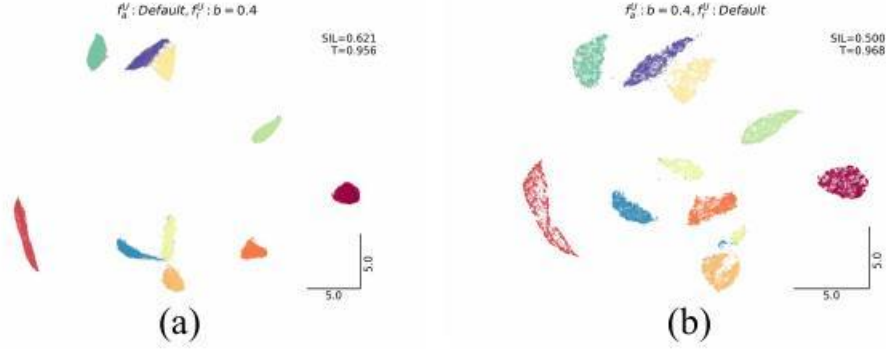


Figure 4: Controlled shape swap on MNIST. (a) Default attraction with the $b = 0.4$ repulsion shape preserves the default UMAP clusters but increases compactness. (b) Default repulsion with the $b = 0.4$ attraction shape creates additional within-cluster structure and new clusters, highlighting attraction as the main driver of cluster formation.

Proposition 6.1. *Neg-t-SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

With $a = 1$ and $b = 1$, UMAP has attraction shape $f_a^U = -2/(1 + \zeta^2)$ and repulsion shape $f_r^U = 2/(\zeta^2(1 + \zeta^2))$. The attraction shape is bounded in $[-2, 0]$ with $f_a^U(0) = -2$ (Fig. 1(f)), but the repulsion shape remains unbounded as $\zeta \rightarrow 0$ (Fig. 1(g)). Because $f_a^U < -1$ near $\zeta = 0$, Proposition 4.1 and Section 5 imply that even this unity-parameter UMAP still needs annealing; Fig. 1(c,d) shows this contrast between annealed and fixed-rate UMAP embeddings.

Damrich et al. (2023) analyzed UMAP’s negative-sampling loss through contrastive embedding (CE) and identified an effective UMAP kernel $1/\zeta^2$. NEG-t-SNE changes that kernel to $1/(1 + \zeta^2)$, which in UMAP notation gives the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$. The resulting shapes are $f_a^N = -2/(2 + \zeta^2)$ and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$ (Figs. 1(f,g)). The attraction shape lies in $[-1, 0]$, so any $\lambda \in [0, 1]$ gives contraction without oscillating between expansion and contraction. The repulsion shape is also bounded in $[0, 1]$ and does not diverge as $\zeta \rightarrow 0$, reducing the chance that randomly sampled repulsive updates eject points from clusters. Previous explanation emphasized bounded repulsion in NEG-t-SNE versus unbounded repulsion in UMAP (Damrich et al. 2023); our contribution, together with the controlled evidence in Fig. 4, shows that bounded attraction is equally important for stability and compactness. Appendix F illustrates this comparison, and Appendix J uses the same reasoning to mix attraction and repulsion shapes across UMAP, NEG-t-SNE, and PaCMAP.

7 Discussion and Conclusion

This paper contributes a force-shape account of cluster formation in neighbor embeddings. The usual summary—attraction pulls similar points together and repulsion pushes dissimilar points apart—is correct but incomplete. By analyzing the distance-dependent shapes of those forces, we identify when attraction contracts or expands, when repulsion controls compactness, and why common optimization heuristics matter. The main text focuses on UMAP and NEG-t-SNE, while Appendix G shows how the same perspective extends to other dimensionality reduction methods.

First, we treated attraction and repulsion coefficients as functional mappings and derived contraction/expansion criteria for their shapes (Propositions 4.1 and 4.2). This revealed a counterintuitive property of UMAP: at short distances, the attractive update can expand rather than contract distances (Fig. 1(b)). UMAP’s learning-rate schedule alleviates this issue by driving the minimum contraction distance toward zero. This same claim anchors the UMAP analysis in Section 5, the NEG-t-SNE comparison in Section 6, and the shape summaries in Fig. 1(b), Fig. 1(f), and Fig. 1(g), all through Proposition 4.1.

Second, we used the same formalism to explain initialization sensitivity. Similar points that start far apart experience weak long-range attraction and may never contract sufficiently. Adding a long-range attraction term improves consistency under random initialization (Fig. 2), suggesting that UMAP can encode a stable structure even when the default dynamics do not always reach it.

Third, we separated the role of repulsion from attraction. Increasing repulsion enlarges inter-cluster distances and compactness (Fig. 3); adding a small positive offset to the repulsion shape gives a form of control that is not available by only varying the minimum-distance parameter m_d .

Fourth, we clarified the relationship between NEG- t -SNE and UMAP. When $a = 1$ and $b = 1$, NEG- t -SNE has bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) shapes, so the contraction and expansion criteria remain stable across fixed learning rates in $[0, 1]$. Allowing a and b to vary reintroduces numerical challenges similar to UMAP. We also showed that NEG- t -SNE and Parametric UMAP are equivalent under the modified loss.

Together, these contributions make attraction–repulsion dynamics easier to inspect, compare, and modify. Because dimensionality reduction and contrastive learning are closely connected (Damrich and Hamprecht, 2021; Hu et al., 2023), the same shape-level reasoning may also help analyze representation learning objectives. The broader value of the work is therefore methodological: it provides a concise diagnostic language for making embeddings more interpretable, consistent, and reliable without changing the underlying scientific claims of the algorithms.

Software and Data

All datasets used to support the paper’s empirical contributions are publicly available. The MNIST dataset is available at yann.lecun.com

Fashion-MNIST is available at github.com.

Single-cell transcriptomes data is available at github.com

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Appendix

The appendix supports the main contribution structure. Appendix [A](#) gives the derivations behind the attraction and repulsion shapes. Appendix [B](#) expands the learning-rate analysis, and Appendix [C](#) applies the attraction/repulsion insight to large language model embeddings. Appendix [D.1](#) defines the Procrustes distance and matrix used to assess initialization consistency, while Appendix [E](#) repeats that experiment on additional datasets. Appendix [F](#) extends the NEG-*t*-SNE/UMAP comparison, Appendix [G](#) applies the shape lens to other dimensionality reduction algorithms, and Appendix [H](#) summarizes high-dimensional graph construction. Appendix [I](#) gives detailed λ_a and λ_r sweeps from Fig. [I](#). Appendix [J](#) shows mix-and-match shape experiments, and Appendix [K](#) records implementation details.

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. To justify the contraction criterion used throughout the paper, we subtract Eq. [\(7\)](#) from Eq. [\(8\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

The distance contracts exactly when $|1 + 2\lambda f_a| < 1$, which gives

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. To justify the expansion criterion for negative edges, we subtract Eq. [\(9\)](#) from Eq. [\(10\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

Since f_r is defined as a nonnegative repulsion shape, the latter condition suffices. □

A.3 Attraction Shape:

We derive the attraction shape from a general low-dimensional affinity, $q_{ij} = (\gamma + a\|y_i - y_j\|_1^{2b})^{-1}$, so the same calculation supports the UMAP case ($\gamma = 1$) and the NEG-*t*-SNE case ($\gamma = 2$). The attractive force is

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

With $\zeta = \|y_i - y_j\|_2$, the scalar coefficient gives the attraction shape

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

The monotonicity assumption used in the main text follows from the derivative of f_a^U :

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

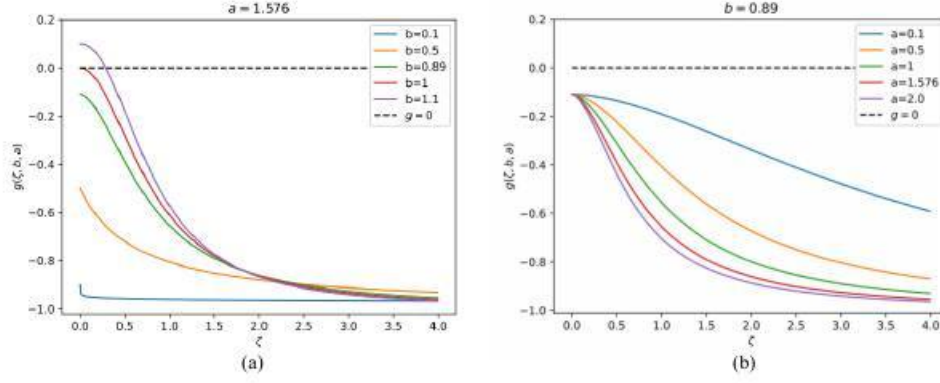


Figure 5: Monotonicity diagnostic for the attraction shape. Values of $g(\zeta, b, a)$ are shown for (a) a fixed at 1.576 and (b) b fixed at 0.89.

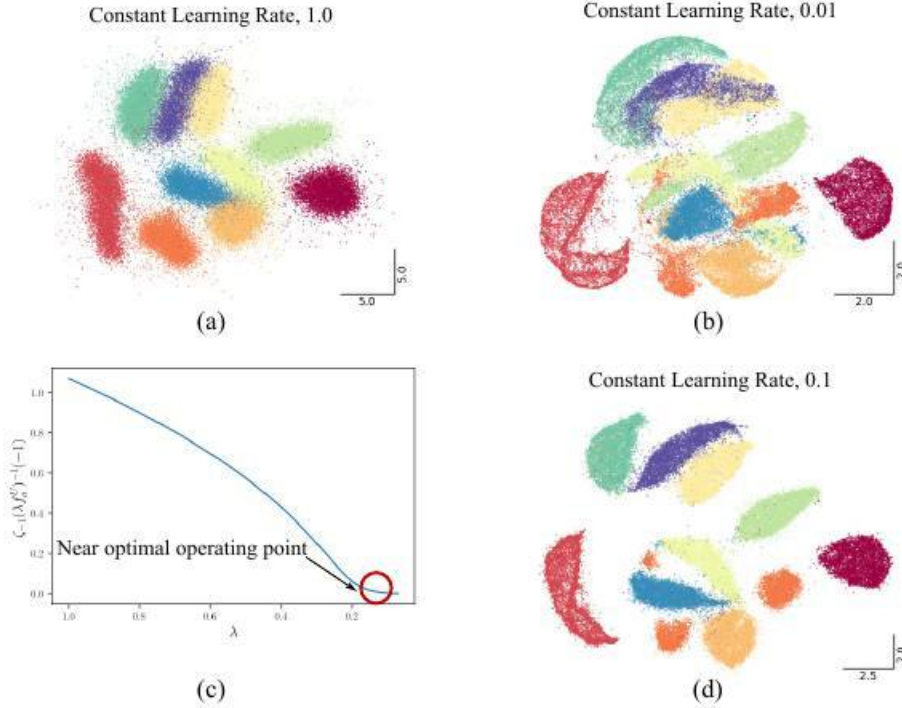


Figure 6: Learning-rate interpretation through ζ_{-1} . (a) A high learning rate ($\lambda = 1.0$) leaves ζ_{-1} large and produces diffuse embeddings. (b) A very low learning rate ($\lambda = 0.01$) weakens both attraction and repulsion, preventing cluster formation. (c) ζ_{-1} decreases nonlinearly with λ , revealing the tradeoff between contraction distance and force strength. (d) A near-optimal constant learning rate $\lambda = 0.1$ produces distinct, compact clusters.

A strictly increasing attraction shape ($\frac{df_a^U}{d\zeta} > 0$) therefore requires

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. The inequality holds for $0 < b \leq 1$ by the derivative and asymptotes of g . Figure 5 visualizes this condition across b and a ; when $b > 1$, inequality (23) no longer holds.

A.5 Repulsion Shape:

The repulsive force, using the general form of the low-dimensional affinity, is given by

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The scalar coefficient gives the repulsion shape

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

Thus the general repulsion shape is positive, matching the expansion criterion used in the main text.

A.6 Loss functions due to modified attraction and repulsion shape:

The modified attraction contribution in Eq. (13) corresponds to the attractive cost

$$-\int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

and the modified repulsion contribution in Eq. (15) corresponds to

$$-\int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases, the added norm term is a regularizer. The shape framing makes the regularizer’s role transparent: the attraction modification increases far-sightedness, while the repulsion modification adds a constant repulsive coefficient.

B More on UMAP’s Learning Rate

Sections 5 and 6 frame learning-rate annealing as a consequence of attraction shape (Fig. 6). The key quantity is the minimum distance for contraction, ζ_{-1} : annealing reduces this value and thereby supports compact clusters. The same lens explains why Neg-*t*-SNE tolerates constant learning rate 1.0 better than UMAP when $a = 1.0$ and $b = 1.0$: Neg-*t*-SNE’s attraction shape lies in $[-1, 0]$, while UMAP’s lies in $[-2, 0]$. Consistently, UMAP becomes more stable at constant learning rate 0.5 when its effective attraction shape stays within $[-1, 0]$ (Figs. 11f,h).

Annealing remains useful across a wide range of a and b because it eventually pushes ζ_{-1} close to zero. If the learning rate stays too high, ζ_{-1} remains large and the embedding is diffuse (Fig. 6a); if it starts too low, both attraction and repulsion are too weak to form clear clusters (Fig. 6b). The ζ_{-1} - λ curve identifies $\lambda = 0.1$ as a near-optimal constant value, with small ζ_{-1} but still substantial force strength (Fig. 6c). At this constant learning rate, clusters are compact and have clear boundaries (Fig. 6d).

C Large Language Model Embedding

This appendix example shows that the attraction/repulsion contribution is not limited to image benchmarks. We randomly selected 300,000 PubMedBERT (Gu et al., 2021) feature vectors with known labels from the PubMed dataset (González-Márquez et al., 2024). Default UMAP gives a crowded map with overlapping labels (Fig. 7a), making the structure difficult to interpret without colors. Reducing b to 0.6, as motivated by Section 5.2, increases repulsion while also strengthening attraction-driven cluster formation. The resulting map is easier to explore even without explicit labels (Fig. 7b). Quantitatively, on a hold-out set of 10,000 points, k-NN accuracy ($k = 10$) increases from 49.2% to 55.48% when b is reduced to 0.6.

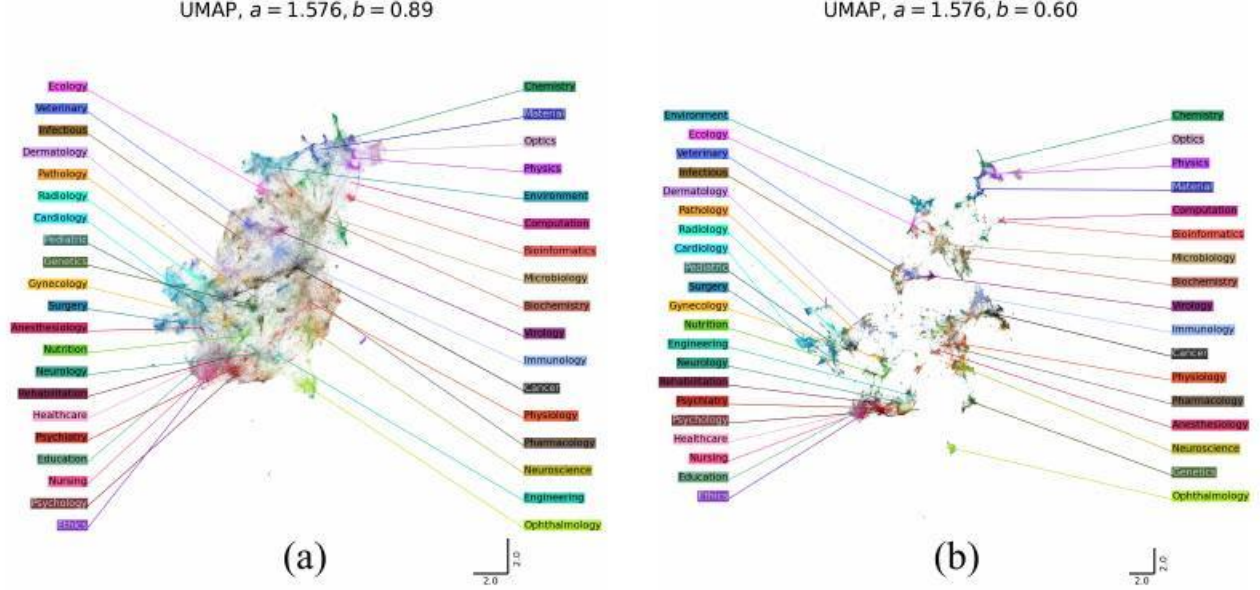


Figure 7: Large-language-model embedding example on 300K PubMed abstracts. (a) Default UMAP ($a = 1.576$, $b = 0.89$) produces overlapping labels. (b) The shape-motivated setting $a = 1.576$, $b = 0.60$ increases repulsion and attraction, yielding clearer cluster formation and less label overlap. Both maps are labeled for interpretability.

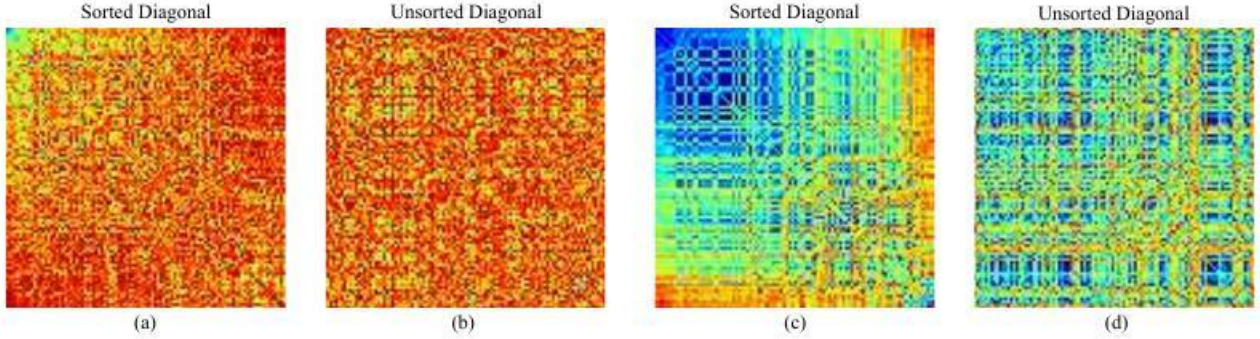


Figure 8: Procrustes-matrix visualization for the initialization-consistency analysis. (a) Procrustes matrix from [2]f for default UMAP, with the diagonal sorted by p_d to the PCA-initialized reference. (b) The same values as in (a), but with an unsorted diagonal. (c) Procrustes matrix from [2]g for the modified attraction shape, again sorted by p_d to the reference. (d) The same values as in (c), but unsorted. Sorting makes run-to-run self-similarity visually apparent.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

Procrustes distance (Gower 1975) supports the initialization-consistency contribution by measuring similarity between two point clouds $\{x\}$ and $\{y\}$ after translation, scaling, and rotation. Operationally, we hold $\{x\}$ fixed and transform $\{y\}$ until the two sets are maximally aligned. Let $\{y'\}$ denote the aligned version of $\{y\}$. The distance is

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

This linear alignment measure has been useful in several embedding settings (McInnes et al. 2018; Islam and Fleischer 2022; Kotlov et al. 2024).

We use Procrustes distance to quantify whether random initializations converge to the same embedding structure. Let $\{x\}_p$ be a reference embedding, obtained with PCA initialization in our experiments, and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ be N embeddings from random initialization. We summarize run-to-run similarity by the mean and

standard deviation of the strictly lower triangular entries of the Procrustes matrix P (reported as mean $\pm$ std in Figs. 9 and 10):

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

For visualization, we sort the indices of X_r so that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$. The diagonal entries then record distances to the reference:

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal entries record pairwise distances among random-initialized embeddings:

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Sorting the diagonal does not change the numerical values, but it makes self-similarity visible. In Fig. 8(a), sorted default-UMAP embeddings that resemble the reference cluster in the upper-left region. With the modified attraction shape, the larger blue region in Fig. 8(c) indicates more mutually similar embeddings and suggests a metastable embedding structure. Without sorting, these regions disappear visually and the consistency signal is much harder to read (Fig. 8(b,d)).

D.2 Trustworthiness

Trustworthiness (Venna and Kaski 2001) quantifies the local-structure side of our evaluations: it measures how well high-dimensional neighborhoods are preserved after dimensionality reduction.

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

Here $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. We use the common choice $k = 5$ to assess preservation of each point's five nearest neighbors. For computational efficiency, reported trustworthiness values use 10,000 randomly sampled indices, with the same indices reused across compared embeddings.

D.3 Silhouette Score

Silhouette score (Rousseeuw 1987) captures the cluster-separation side of our evaluations. Although originally designed for clustering algorithms, we use it to measure how well ground-truth labels separate in an embedding. For a point y_i in point cloud $\{y\}$, let a_i be the mean distance from y_i to points with the same label, and let b_i be the minimum mean distance from y_i to points with any other label. The pointwise silhouette is

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

The value lies in $[-1, 1]$: values near 1 indicate that y_i fits its own label cluster, values near 0 indicate boundary points, and values near -1 indicate failed label separation. The overall score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We compute silhouette score on the full embedding, without random sampling, using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

Section 5.1 presents the initialization-consistency contribution on MNIST. Here we repeat the same experiment on Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) (Figs. 9 and 10). The conclusion is unchanged: modified and composite attraction shapes that increase long-range attraction substantially improve reconstruction consistency.

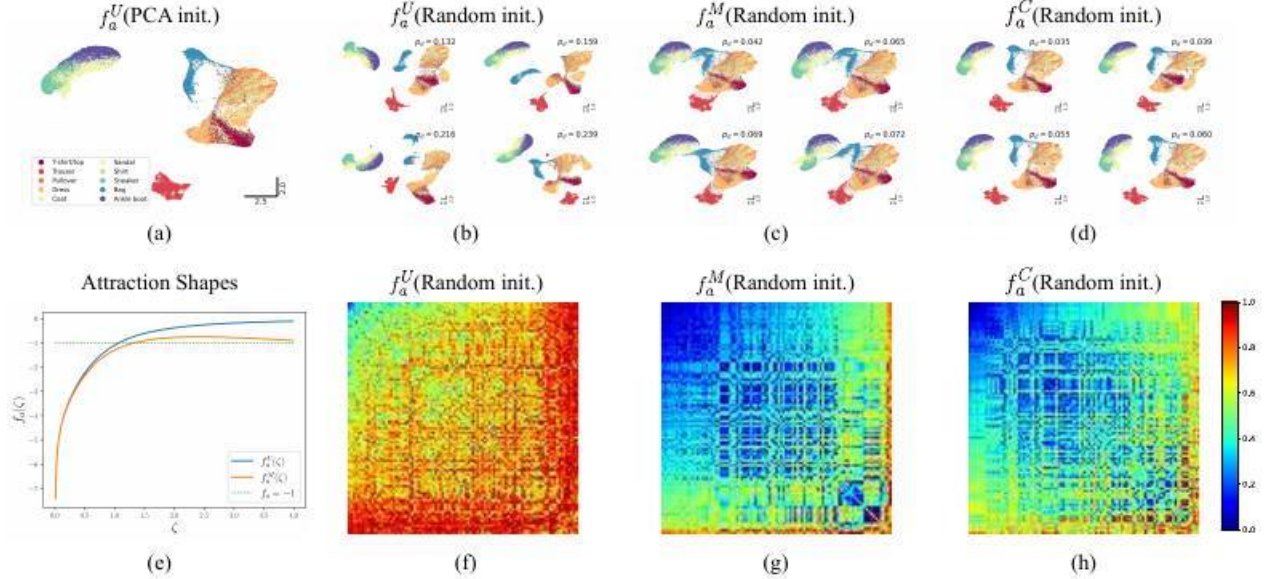


Figure 9: Initialization-consistency contribution on FMNIST. (a) PCA-initialized reference mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) to (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

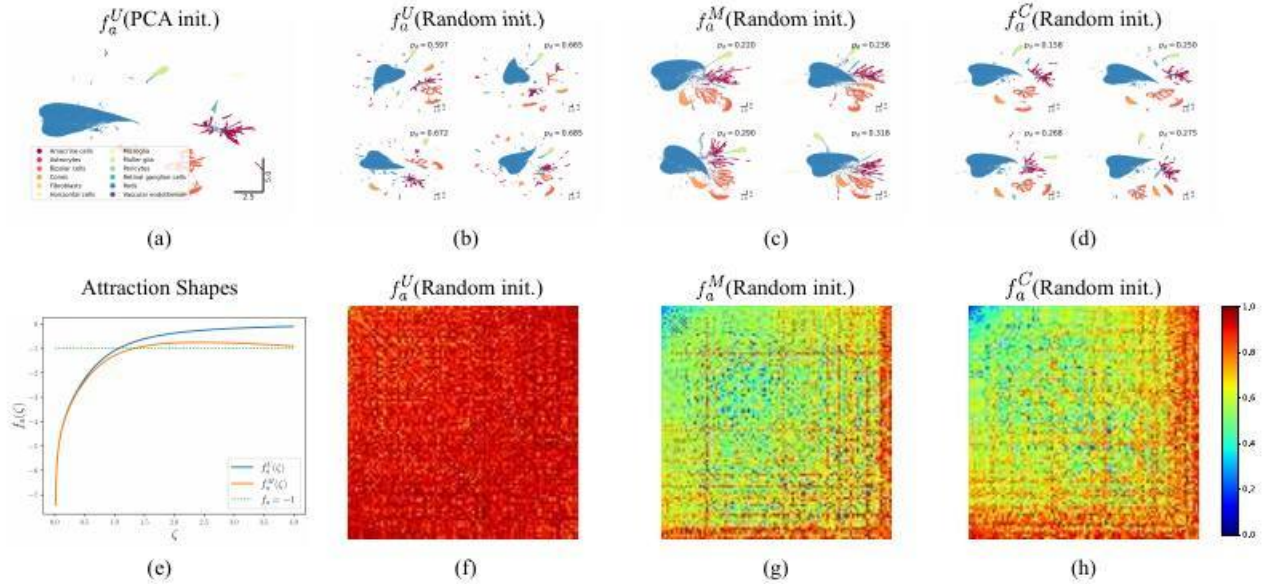


Figure 10: Initialization-consistency contribution on single-cell transcriptomes. (a) PCA-initialized reference mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) to (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

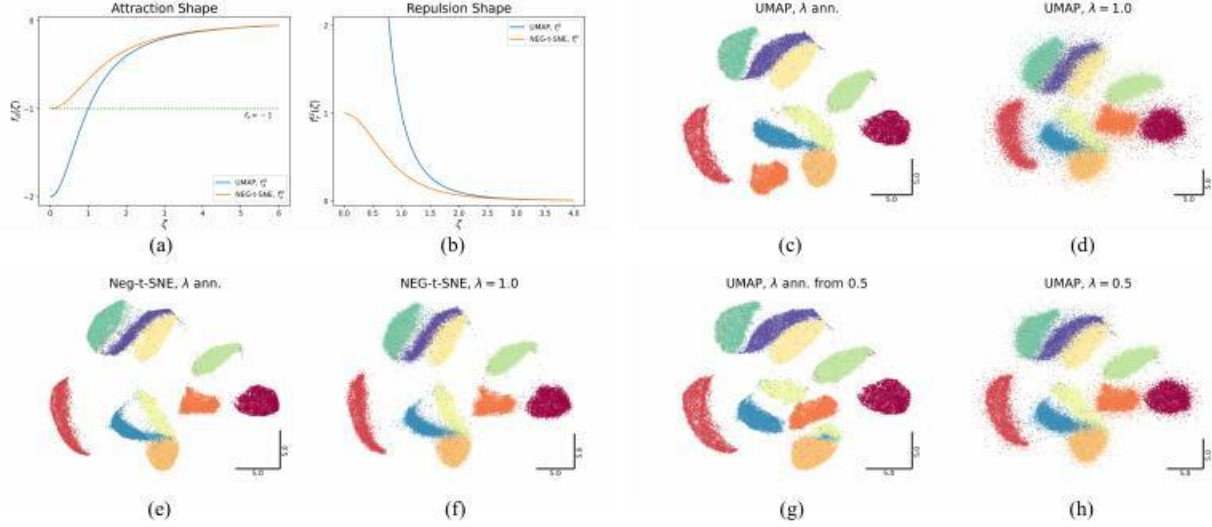


Figure 11: Learning-rate sensitivity explained by attraction shape on MNIST. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP is sensitive to λ because $f_a^U < -1$ at small ζ , so clusters become fuzzy without annealing. (e,f) NEG- t -SNE is less sensitive because $f_a^N \in [-1, 0]$, yielding cleaner clusters even without annealing. (g,h) Setting $\lambda = 0.5$ confines UMAP’s effective attraction to $[-1, 0]$ and reduces sensitivity.

Figure 11 extends the NEG- t -SNE comparison by showing both shapes and MNIST embeddings. With annealing, UMAP and NEG- t -SNE produce similar outputs (Figs. 11(c,e)). With constant learning rate 1, UMAP becomes fuzzy whereas NEG- t -SNE keeps clearer boundaries (Figs. 11(d,f)). Section 6 explains this through attraction: constraining f_a^U to $[-1, 0]$ should reduce fuzziness at fixed λ . This agrees with Fig. 1(h,i), where UMAP improves when the attraction learning rate (λ_a) is $\lesssim 0.5$. Initializing UMAP with $\lambda = 0.5$ satisfies Proposition 4.1 for all ζ , and the resulting embeddings have sharper boundaries than the constant-1 case (Figs. 11(d,g,h)). Some points still remain outside clusters because UMAP’s repulsion shape is unchanged, a limitation addressed by NEG- t -SNE.

Introducing parameters a and b into NEG- t -SNE gives essentially the Parametric UMAP formulation (Proposition 6.1). The affinity becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, with attraction and repulsion shapes $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$ and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$. For $0 < b < 1$, both shapes are unbounded as $\zeta \rightarrow 0$, so varying a and b reintroduces UMAP-like numerical challenges. Compared with UMAP, the attraction shape can achieve a lower minimum contraction distance (ζ_{-1}), which may help cluster formation; however, it also decays faster with distance, increasing near-sightedness and weakening long-range attraction.

F.2 Comparison to Parametric UMAP

Parametric UMAP was initially trained with the original UMAP objective (Sainburg et al., 2021); later work used a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al., 2023). The contribution of this subsection is to show that this modification makes Parametric UMAP equivalent to Neg- t -SNE (Damrich et al., 2023) (Proposition 6.1).

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both these are NEG-t-SNE with explicit parameters a and b (while in Neg-t-SNE these are set to 1). $\square$

G Alternate Dimensionality Reduction Algorithms

This section shows that the paper’s shape framework is not specific to UMAP. The alternative algorithms considered here use the same low-dimensional kernel as UMAP with $a = 1$ and $b = 1$:

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We first discuss TriMap (Amid and Warmuth, 2019). Although TriMap uses triplets rather than pairwise interactions, it provides a useful stress test for shape analysis because attraction and repulsion are coupled. We then discuss PaCMAP (Wang et al., 2021) and LocalMAP (Wang et al., 2025), which adapt TriMap-like ideas to pairwise interactions. Finally, we examine t -SNE (Van der Maaten and Hinton, 2008), SNE (Hinton and Roweis, 2002), and multidimensional scaling (Borg and Groenen, 2007).

G.1 TriMap

TriMap (Amid and Warmuth, 2019) optimizes a low-dimensional embedding with a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} weights triplet (y_i, y_j, y_k) , y_j is a high-dimensional k -nearest neighbor of y_i , and y_k is a far-away point. Minimization should attract y_i and y_j while repelling y_i and y_k . The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the distance between nearest neighbors, $\zeta_2 = \|y_i - y_k\|_2$ is the distance from the faraway point, and f_a^T and f_r^T are attraction and repulsion shapes of TriMap, respectively. Unlike UMAP, the attractive and repulsive components are non-separable and the shapes depend on two distance measures (making them 2D). The functional form of the attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

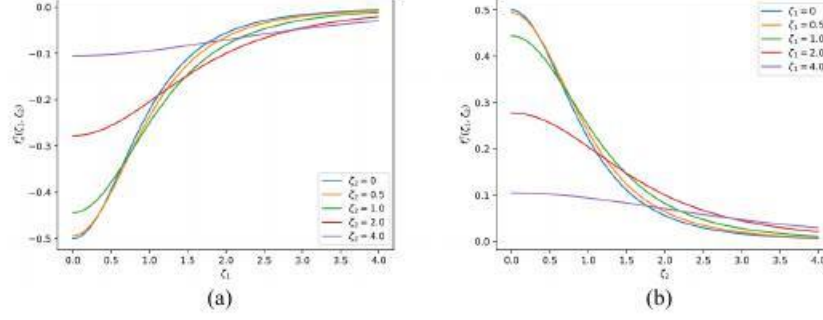


Figure 12: TriMap shape diagnostics. (a) Attraction shapes for different ζ_2 values and (b) repulsion shapes for different ζ_1 values.

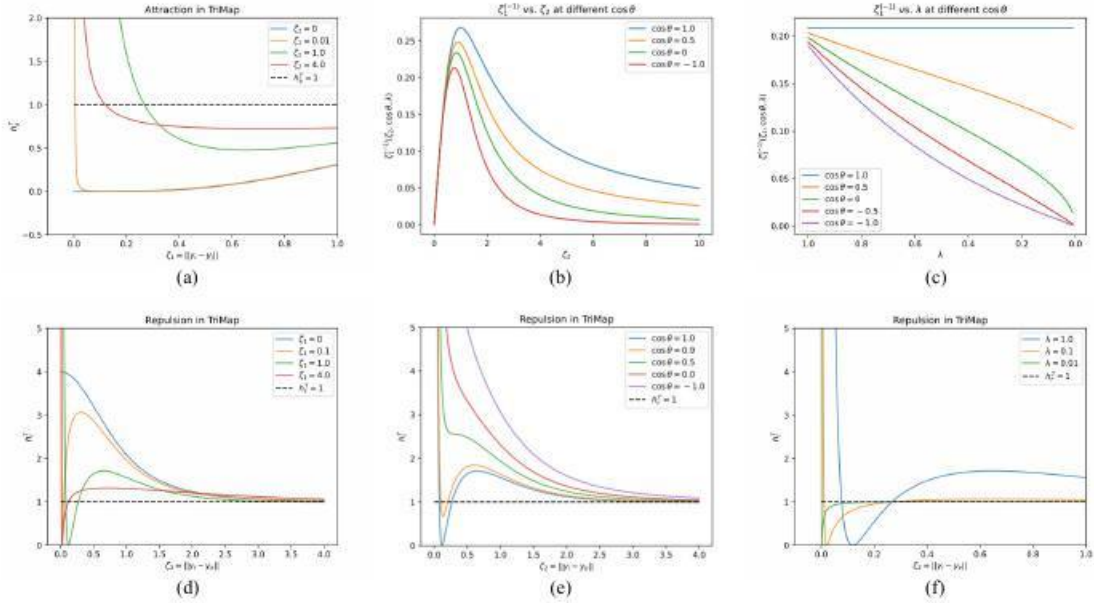


Figure 13: TriMap contraction and expansion conditions. (a) h_a^T versus ζ_1 for different ζ_2 values; values below the dotted line indicate attraction. Like UMAP, TriMap can show both attraction and expansion for nearest neighbors (y_i, y_j) . (b,c) The minimum contraction distance ($\zeta_1^{(-1)}$) varies with (b) ζ_2 and (c) λ , with $\cos \theta$ regulating the range. (d-f) Repulsion behavior under changes in (d) ζ_1 , (e) $\cos \theta$, and (f) λ . Values above (below) the dotted line indicate repulsion (attraction). Unlike UMAP's one-sided repulsion shape, TriMap's repulsive term can also contract. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

TriMap’s shapes (Fig. 12) resemble UMAP’s trends, but the minimum attraction value is -0.5 , so attraction alone does not flip point positions as in UMAP. However, Eqs. (45)-(47) couple attraction and repulsion, so Propositions 4.1 and 4.2 do not apply directly. We therefore derive TriMap-specific conditions, beginning with attraction:

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends on ζ_1 , ζ_2 , $\cos \theta$, and λ , so the contraction distance $\zeta_1 = \|y_i - y_j\|_2$ is coupled to additional geometry. Figure 13(a) shows attraction behavior for different ζ_2 with $\cos \theta = 1$ and $\lambda = 1$: values below the dotted line contract ζ_1 , while values above expand it. The intersection with the dotted line defines the minimum contraction distance $\zeta_1^{(-1)}$, analogous to UMAP’s ζ_{-1} :

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

For most cases, $\zeta_1^{(-1)}$ is finite and positive (Fig. 13(b,c)). TriMap can therefore exhibit a UMAP-like need for annealing or a related adaptive strategy; the implementation uses delta-bar-delta (Jacobs, 1988) with appropriate initialization.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

The repulsion condition in Eq. (54) also depends on ζ_1 , ζ_2 , $\cos \theta$, and λ . Here the target is expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) vary the other quantities: values above the dotted line expand distance, whereas values below contract it. Unlike UMAP, TriMap’s repulsive term can therefore create contraction. Because this occurs at small distances, initialization and triplet choice can avoid the anomaly.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) provides another test of the contribution because it optimizes low-dimensional structure at multiple scales. Its loss is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} weight nearest-neighbor (NN), mid-near (MN), and further-pair (FP) interactions (details in Appendix H). The first two terms are attractive and the last is repulsive. Viewed through shapes, PaCMAP is a separable modification of TriMap’s triplet loss: it replaces TriMap’s distant-point affinity with constants (1/10 for

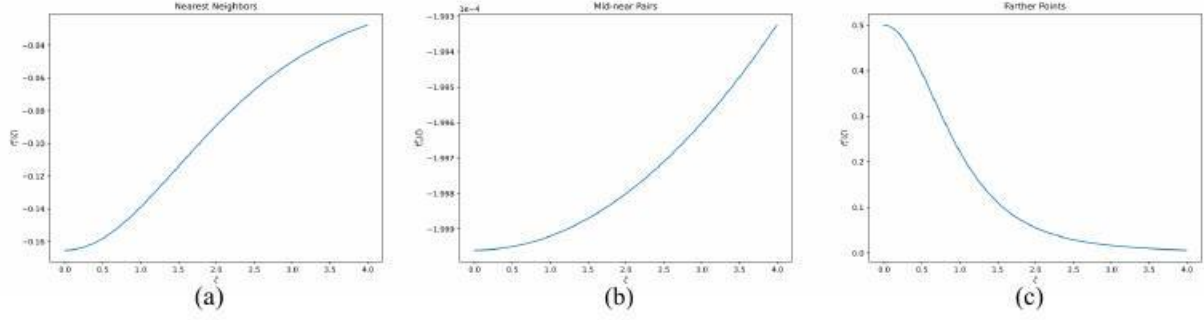


Figure 14: PaCMAP through the shape framework. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near points, with mid-near values on the order of 10^{-4} . (c) Repulsion shape for farther pairs. All panels use $\lambda = 1$.

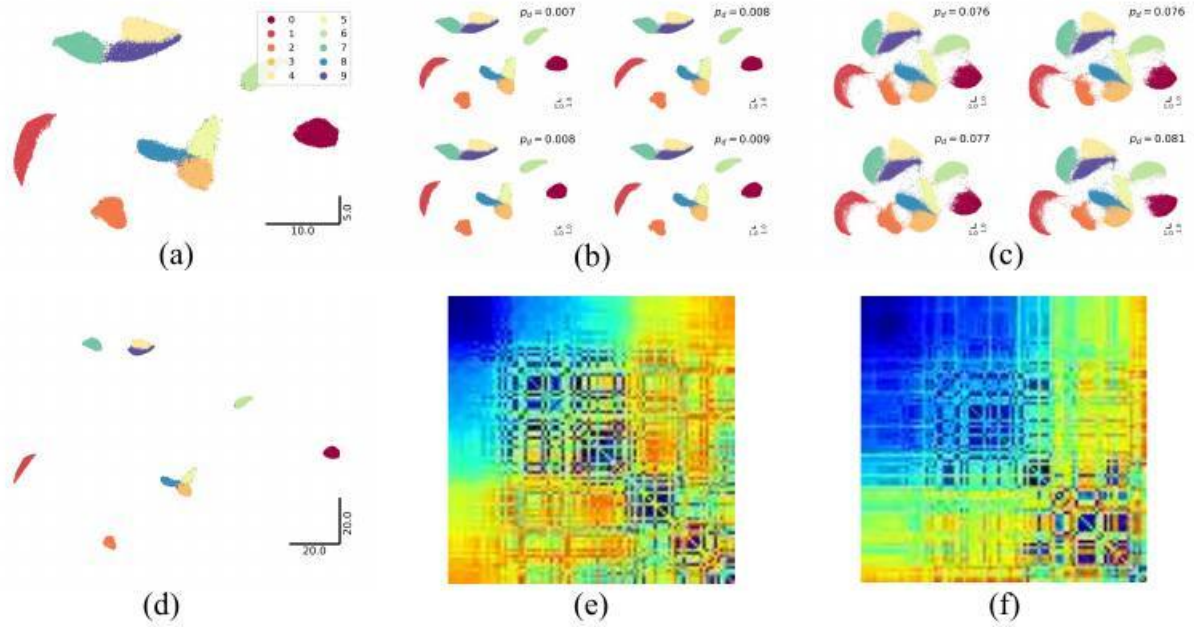


Figure 15: PaCMAP responds to the same shape interventions as UMAP. (a) PCA-initialized PaCMAP embedding of MNIST. (b,c) Four random-initialized samples closest to (a) for (b) original PaCMAP and (c) PaCMAP with modified attraction $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP with modified repulsion $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP (0.51 ± 0.22) and (f) modified attraction (0.43 ± 0.22). Increased long-range attraction improves consistency, while increased repulsion yields smaller clusters and larger inter-cluster distances.

nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther pairs). The update equations therefore decouple. For nearest neighbors, the updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

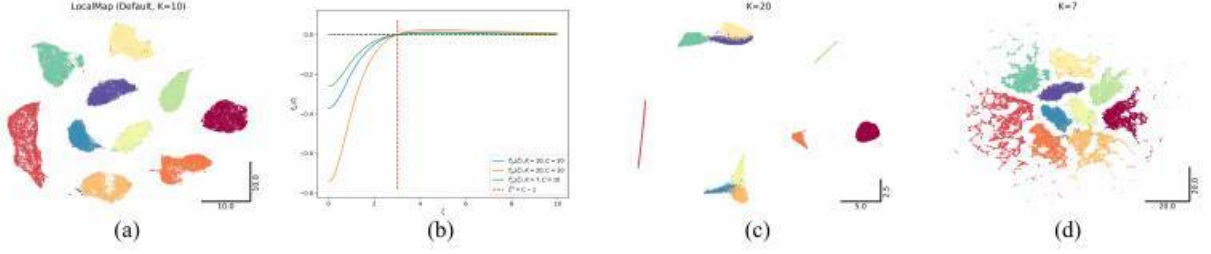


Figure 16: LocalMAP illustrates the attraction–repulsion tradeoff on MNIST. (a) Default embedding. (b) Nearest-neighbor attraction–repulsion shape; values below (above) the dotted line indicate attraction (repulsion), with the transition at $\zeta = \sqrt{C} - 1$. Following LocalMAP, we use $C = 10$. (c) Large K ($= 20$) makes repulsion dominate and compacts clusters. (d) Small K ($= 7$) makes attraction dominate and breaks clusters apart.

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, $\zeta_3 = \|y_i - y_l\|_2$ are distances, f_a^P and f_m^P are attraction shapes for nearest neighbors and mid-near pairs, respectively, and f_r^P is the repulsion shape for the farthest pairs. Correspondingly, the functional forms of the shapes are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

The shapes f_a^P and f_m^P satisfy Proposition 4.1 and f_r satisfies Proposition 4.2 (Fig. 14). PaCMAP attraction is weaker than UMAP attraction, but it remains effective over a broad learning-rate range under Adam (Kingma and Ba, 2015); with $w_{NB} = 3$, maximum attraction stays below 0.5, avoiding flips during attractive updates. PaCMAP typically initializes points in a small low-dimensional sphere, often on the order of 10^{-3} (Wang et al., 2021), and uses repulsion to separate clusters. It largely recovers UMAP-like clustering on MNIST with improved ordering. Its random-initialization consistency is better than UMAP’s (Fig. 15b,e), improves further with modified attraction (Fig. 15c,f), and becomes more compact when repulsion is increased.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al., 2025) extends PaCMAP and provides a clear example of the attraction–repulsion tradeoff. It separates all 10 MNIST clusters (Fig. 16a), with behavior resembling the compactness and splitting effects in Figs. 3g,h,j,k) and 4a,b). We therefore use LocalMAP to examine how the same pair can switch between attraction and repulsion.

LocalMAP first runs PaCMAP and then performs additional optimization to decouple clusters. It minimizes

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term combines attraction and repulsion on the same nearest-neighbor pair: one distance regime contracts, and another expands. The second term is PaCMAP’s farther-pair repulsion, with periodic resampling. We therefore analyze the nearest-neighbor term, whose updates are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction–repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C - 1 - \zeta^2)}{2\sqrt{1 + \zeta^2}(1 + C + \zeta^2)^2}. \quad (69)$$

The update contracts when $\zeta^2 < C - 1$ and $-1 < \lambda f_{ar}^L < 0$ (Proposition 4.1). When $\zeta^2 > C - 1$, $f_{ar} > 0$, so Proposition 4.2 predicts expansion. Thus K and C determine whether attraction or repulsion dominates. In the

LocalMAP implementation, $C = 10$ and $K = 10$ (Fig. 16(b)), and attraction is stronger than in PaCMAP (Fig. 14(a)). For $\zeta > 3$, nearest-neighbor pairs experience repulsion, which separates bridge pairs between clusters; K scales the strength of this separation.

Proposition 4.1 also bounds K : requiring $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the largest attraction that avoids flipping pair placements, which helps preserve PaCMAP's ordering before LocalMAP separation. Equation (69) simplifies to $K \leq \frac{\lambda(1+C)^2}{C-1}$, giving $K \lesssim 13.44$ for $C = 10$ and $\lambda = 1$. When $K \gtrsim 13.44$, repulsion dominates and clusters become compact, but LocalMAP's goal can fail because the bridge between clusters persists (Fig. 16(c), $K = 20$). When K is smaller, attraction dominates because repulsion is too weak, and existing clusters break apart (Fig. 16(d), $K = 7$), matching the attraction-driven splitting in Fig. 4(b).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton, 2008) fits naturally into the shape framework because it also optimizes pairwise distances. Its loss is

$$\mathcal{L} = - \sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the pair weight. Unlike UMAP, the original t -SNE considers all pairs rather than only nearest neighbors. The loss decomposes into attraction and repulsion as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term is attractive and the second is repulsive. The attractive term matches UMAP with $a = 1$ and $b = 1$, but the repulsive term couples all pairs and is therefore costly. Applying the same shape principles used for UMAP, and retaining the all-pairs weights $w_{i,j}$, gives the t -SNE update equations. The attractive update equations are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equation for the repulsive parts are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$. The functional forms of these shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1 + \zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1 + \zeta^2)^2}. \quad (77)$$

For the attractive updates in Eqs. (72)-(73), f_a^{t-SNE} obeys Proposition 4.1 after weighting (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$), giving a minimum contraction distance ζ_{-1} . The repulsive update is coupled across all pairs, so a simple one-shape condition is unavailable. Using the same reasoning as in the TriMap analysis, we obtain the following criterion:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k, k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l, l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

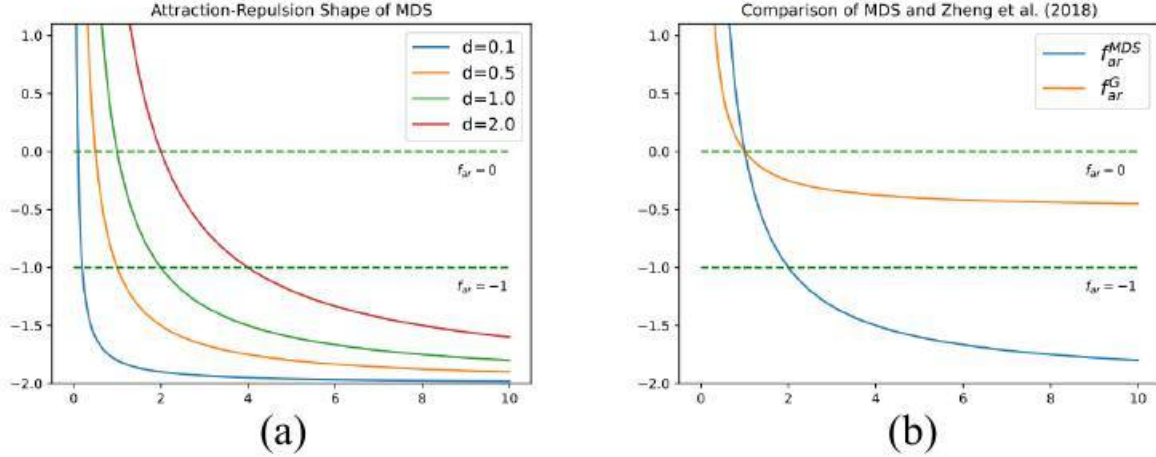


Figure 17: MDS viewed through an attraction-repulsion shape. (a) Shape f_{ar} for MDS. (b) Comparison between the MDS shape and the bounded graph-drawing variant of (Zheng et al., 2018) for $d = 1.0$.

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Because the t -SNE repulsion condition in Eq. (78) has the same structure as TriMap’s condition, its repulsive forces can also create either expansion or contraction depending on the surrounding geometry.

Because t -SNE scales attraction by $w_{i,j}$ and repulsion by $\frac{w_{i,j}}{Z}$, its forces are typically weaker than UMAP’s, motivating large learning rates such as $\approx 10^3$ in Open- t -SNE (Poličar et al., 2024). Most implementations also use early exaggeration, multiplying attractive forces during the first iterations. In shape terms, early exaggeration induces far-sightedness: far-apart neighbors move together, while very close points near $\zeta_{i,j} \approx \zeta_{-1}$ can still separate when needed. This explains why early exaggeration is central for consistent t -SNE embeddings, especially under random initialization. When applied throughout optimization, it makes t -SNE embeddings resemble UMAP embeddings (Böhm et al., 2022).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis, 2002), one of the earliest methods in this family, uses the same loss form as t -SNE (Eq. (71)) but with $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. Its attraction shape is

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which follows Proposition G.3 after replacing the t -SNE symbols with SNE counterparts. This attraction shape depends heavily on learning rate (λ) and pair weights ($w_{i,j}$) for contraction, which contributes to overlapping clusters even for small datasets, the crowding problem (Van der Maaten and Hinton, 2008). t -SNE and later methods mitigate this by replacing the Gaussian kernel with the heavy-tailed kernel in Eq. (43).

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) is usually optimized by stress majorization rather than gradient descent because gradient methods often fail to find good maps. Still, gradient formulations exist (Kruskal, 1964; Zheng et al., 2018), and they are useful for showing the limits of the shape perspective. In particular, Zheng

et al. (2018) gives a gradient-descent MDS method for graph drawing. Ignoring $w_{i,j}$ without loss of generality for that discussion, or absorbing it into the learning rate, the cost is

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss has one term per pair and no separate attraction or repulsion terms; each pair term can act as either. Because the loss is separable across (i, j) components, a single-pair shape analysis is possible. The derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

Thus, the attraction-repulsion shape of MDS becomes

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive for $\zeta > d$ and lies in $[-2, 0]$ in that range (Fig. 17(a)). However, the contraction criterion shows why this is difficult for gradient optimization: only values in $[-1, 0]$ contract, while values outside that range expand; values below -2 flip and expand. To avoid this, Zheng et al. (2018) uses a force-directed-graph-inspired shape

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which works for effective learning rate $\lambda \leq 1$ because f_{ar}^G is bounded in $[-0.5, 0]$. For $\zeta < d$, $f_{ar}^{MDS} > 0$ is repulsive. This interaction works best when initialization is already close to the desired output. Continued pairwise optimization oscillates around $\zeta = d$, where $f_{ar}^G = 0$, giving a distance notion analogous to UMAP's ζ_{-1} . Unlike UMAP, no learning-rate choice drives this distance to zero because preserving it is the MDS objective. Consequently, MDS clusters are often fuzzier than UMAP or PaCMAP clusters (see Lambert et al. (2022), de Bodt et al. (2025), and Kury et al. (2025)). The method of Lambert et al. (2022) modifies MDS with quartet stress and relative distances, enabling global regularization and lower computation than majorization.

H Construction of the High-Dimensional Graph

High-dimensional graph construction supplies the affinities on which the later force shapes operate. SNE Hinton and Roweis (2002) constructs a graph for $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ using

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is chosen to match a user-defined value *perplexity*, P , defined as

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP instead builds the high-dimensional graph from a k-nearest neighbor (k-NN) system:

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k -nearest neighbors of the point x_i and σ_i is a scaling parameter such that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized by a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses the k-NN graph directly, with affinities set to 1, and applies a self-tuning distance measure (Zelnik-Manor and Perona 2004):

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance between x_i and its Euclidean fourth-to-sixth nearest neighbors. Although the definitions differ, this parameter plays the same scaling role as the σ_i terms in Eq. (87) and Eq. (90).

Regardless of graph-construction details, our analysis indicates that the attraction and repulsion shapes of the optimization are the primary drivers of the final low-dimensional geometry.

I Detailed Results for Varying λ_a and λ_b

This section provides the detailed sweeps behind Fig. I(h,i). We obtain them by replacing UMAP’s attraction and repulsion shapes with shapes from other methods while varying λ_a and λ_r . Figure 18 reproduces the main-text MNIST summary. Figures 19-22 show the individual MNIST embeddings for each λ_a and λ_r setting, together with the initialization, an annealed-learning-rate reference, and the cases where either λ_a or λ_r is set to 0. For PaCMAP, we also include the reference with mid-near pairs. Figures 23-27 and Figs. 28-32 provide the corresponding FMNIST and single-cell transcriptomes results.

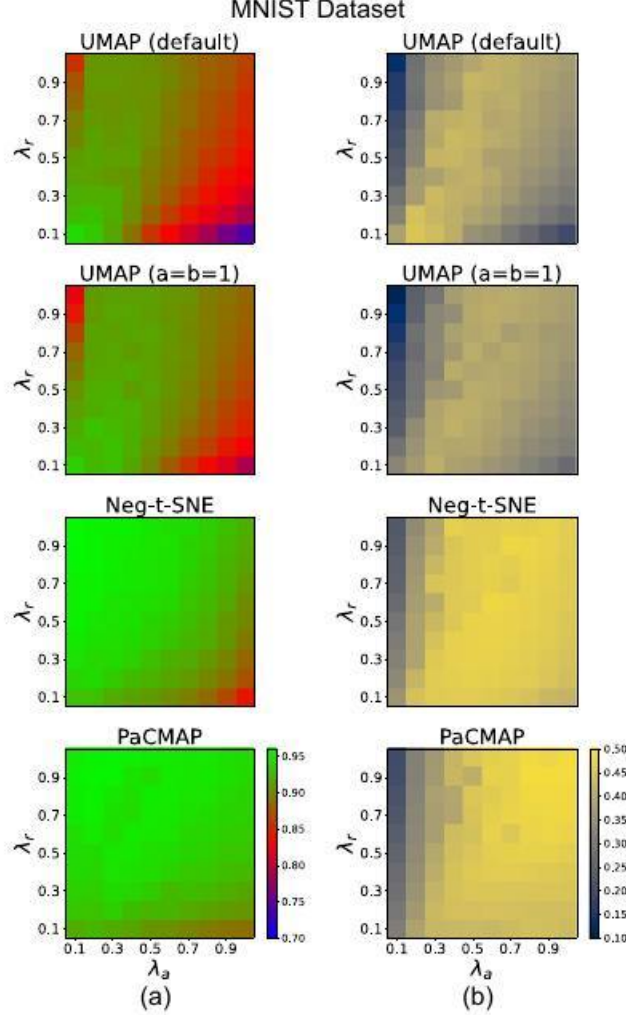


Figure 18: MNIST shape-sweep summary. (a) Trustworthiness and (b) Silhouette score for different methods under the attraction/repulsion learning-rate sweep.

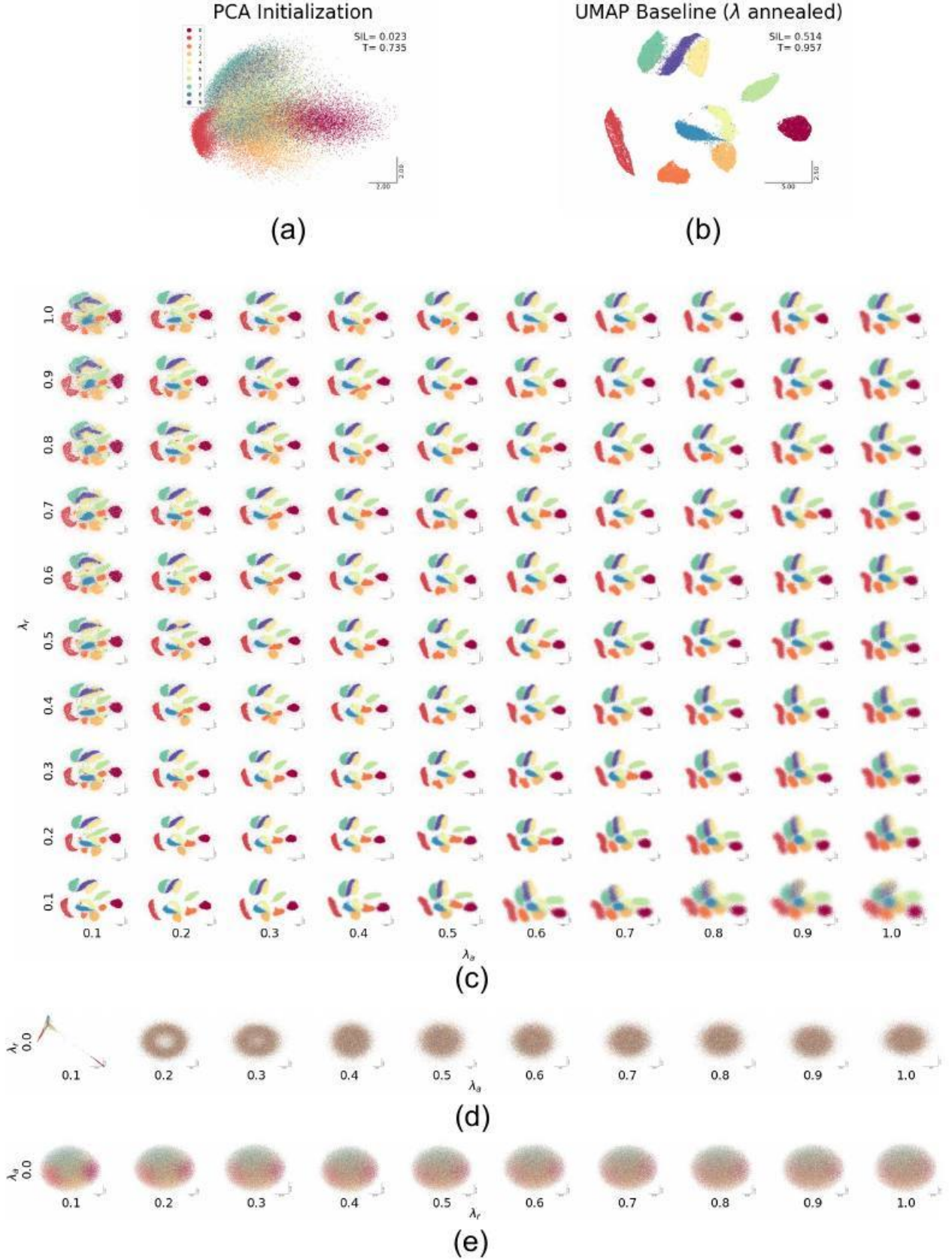


Figure 19: MNIST detailed shape sweep varying λ_a and λ_r with UMAP’s attraction and repulsion shapes. (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

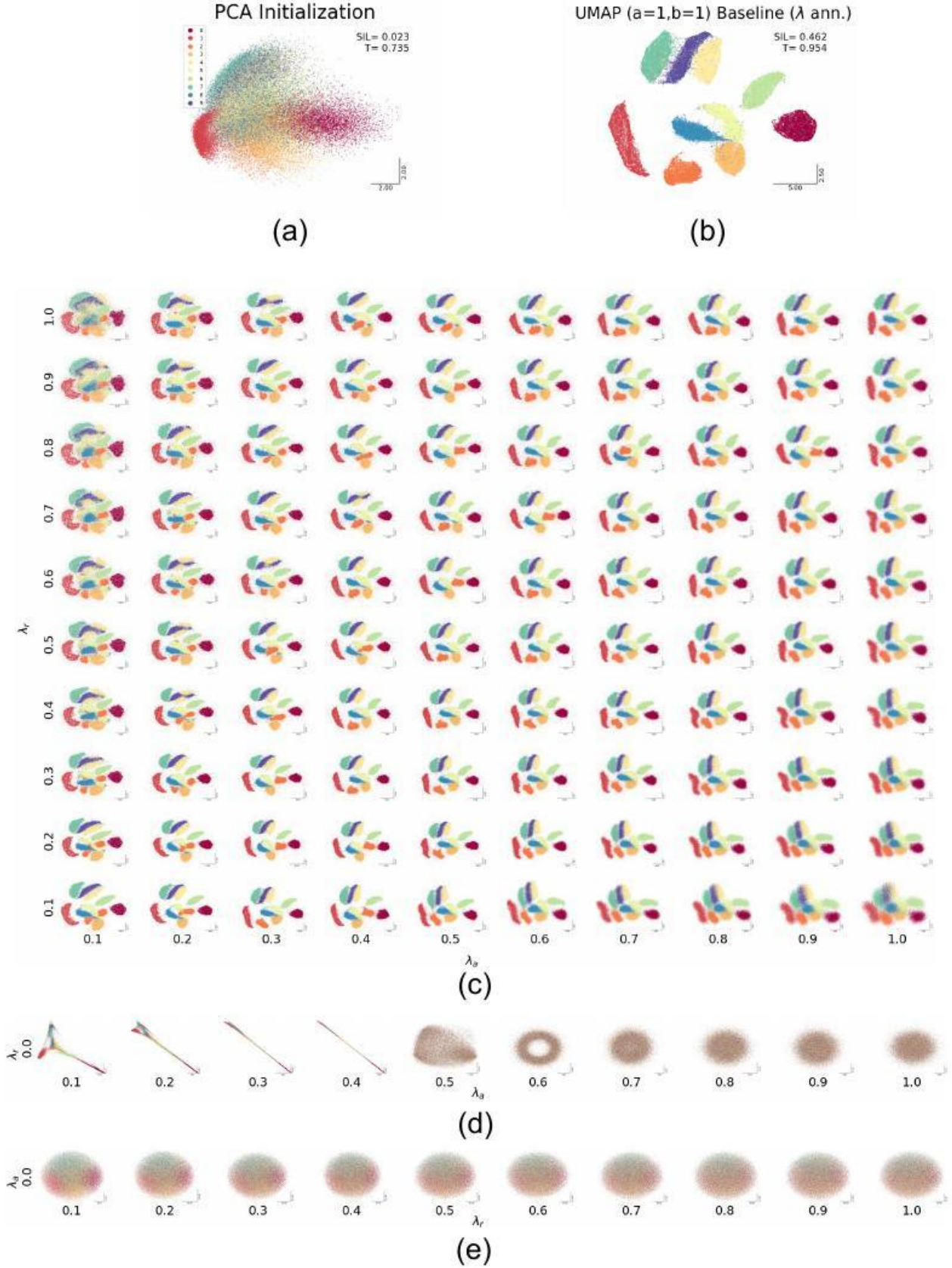


Figure 20: MNIST detailed shape sweep varying λ_a and λ_r with UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

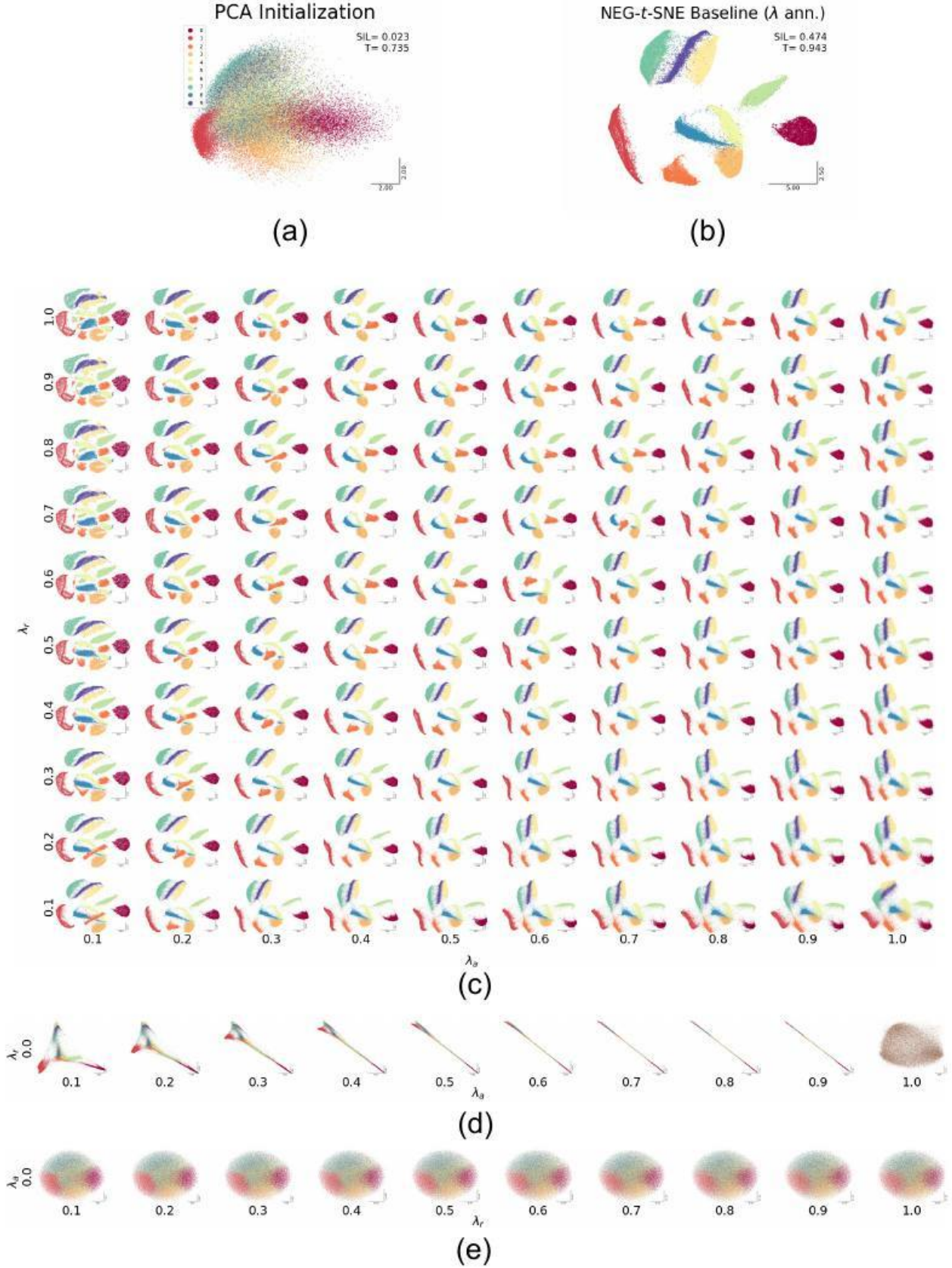


Figure 21: MNIST detailed shape sweep varying λ_a and λ_r with NEG-t-SNE’s attraction and repulsion shapes. (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

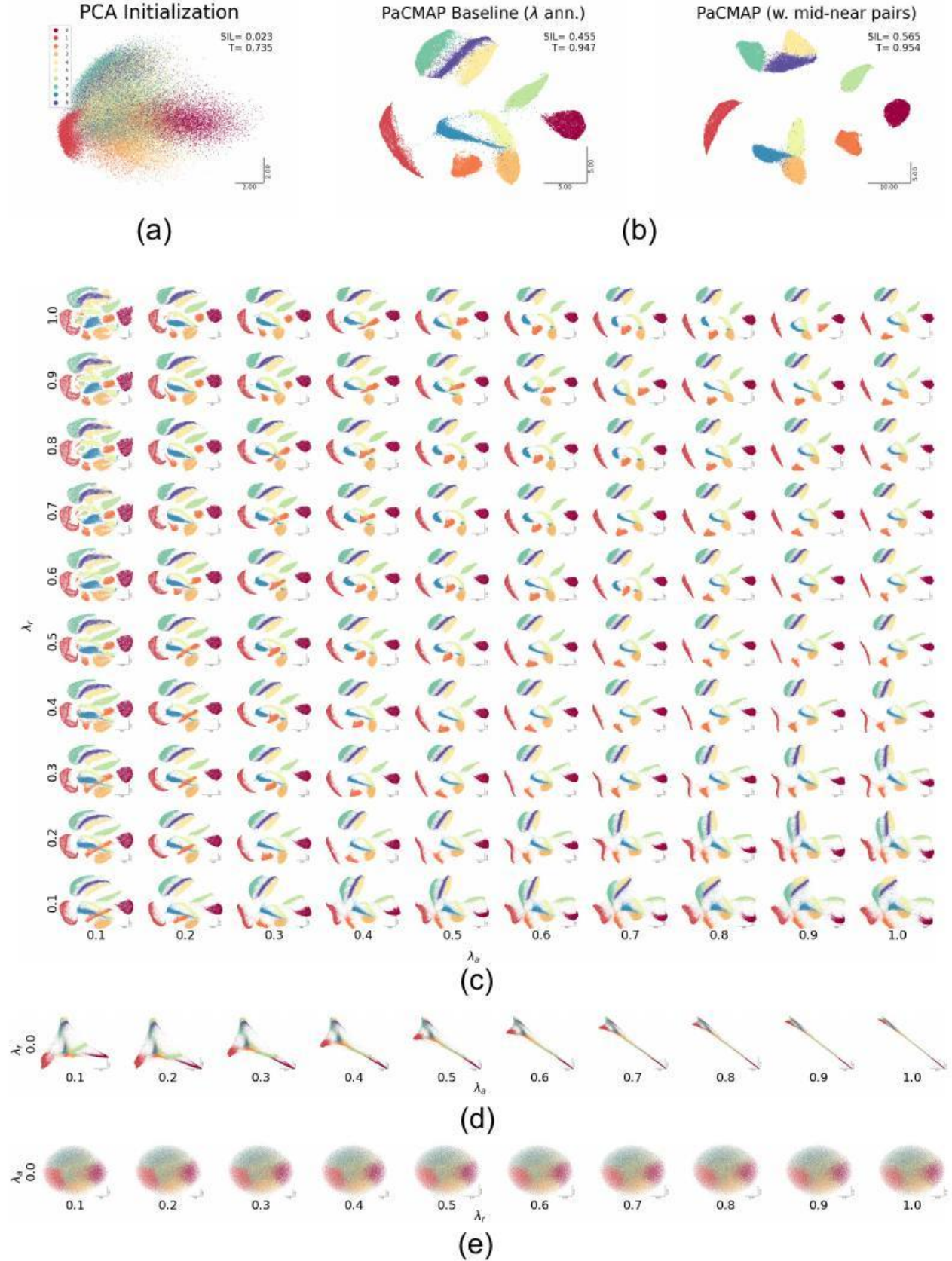


Figure 22: MNIST detailed shape sweep varying λ_a and λ_r with PaCMAP’s attraction and repulsion shapes. (a) Initialization. (b) Left: annealed- λ baseline without mid-near points to isolate attraction–repulsion; right: baseline with mid-near points included. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

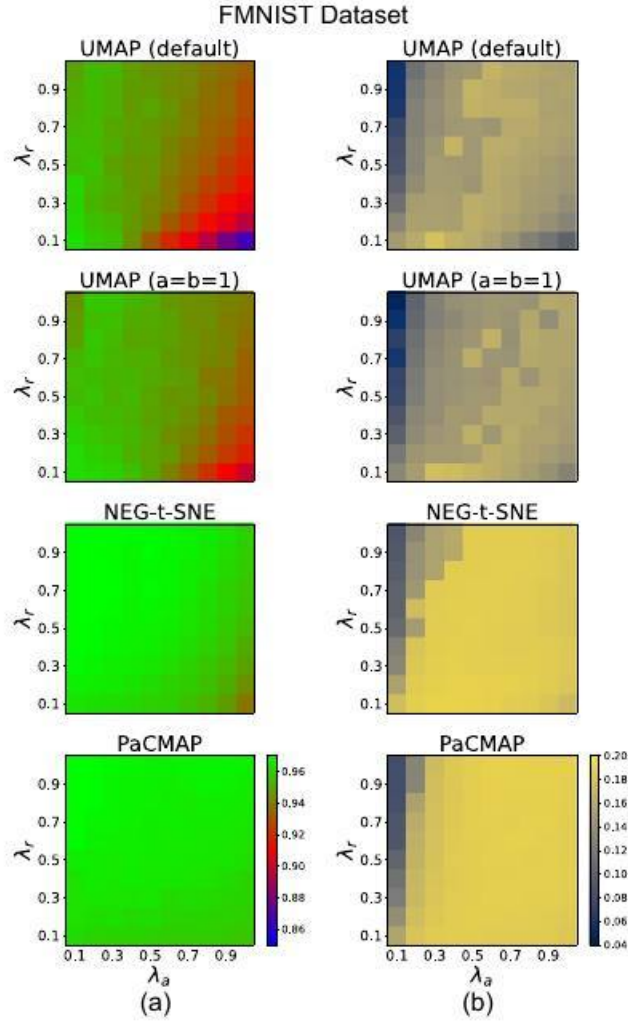


Figure 23: FMNIST shape-sweep summary. (a) Trustworthiness and (b) Silhouette score for different methods under the attraction/repulsion learning-rate sweep.

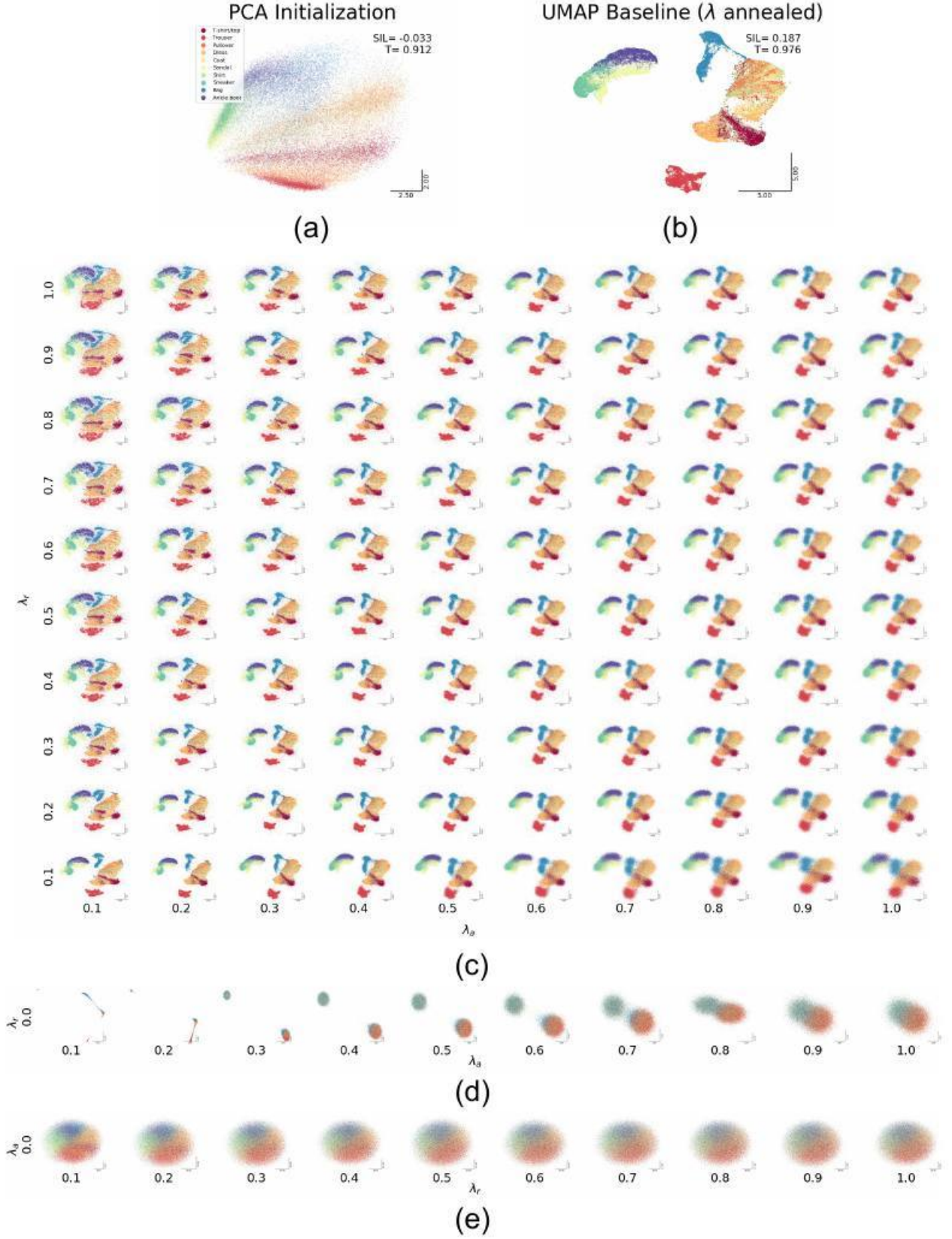


Figure 24: FMNIST detailed shape sweep varying λ_a and λ_r with UMAP’s attraction and repulsion shapes. (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

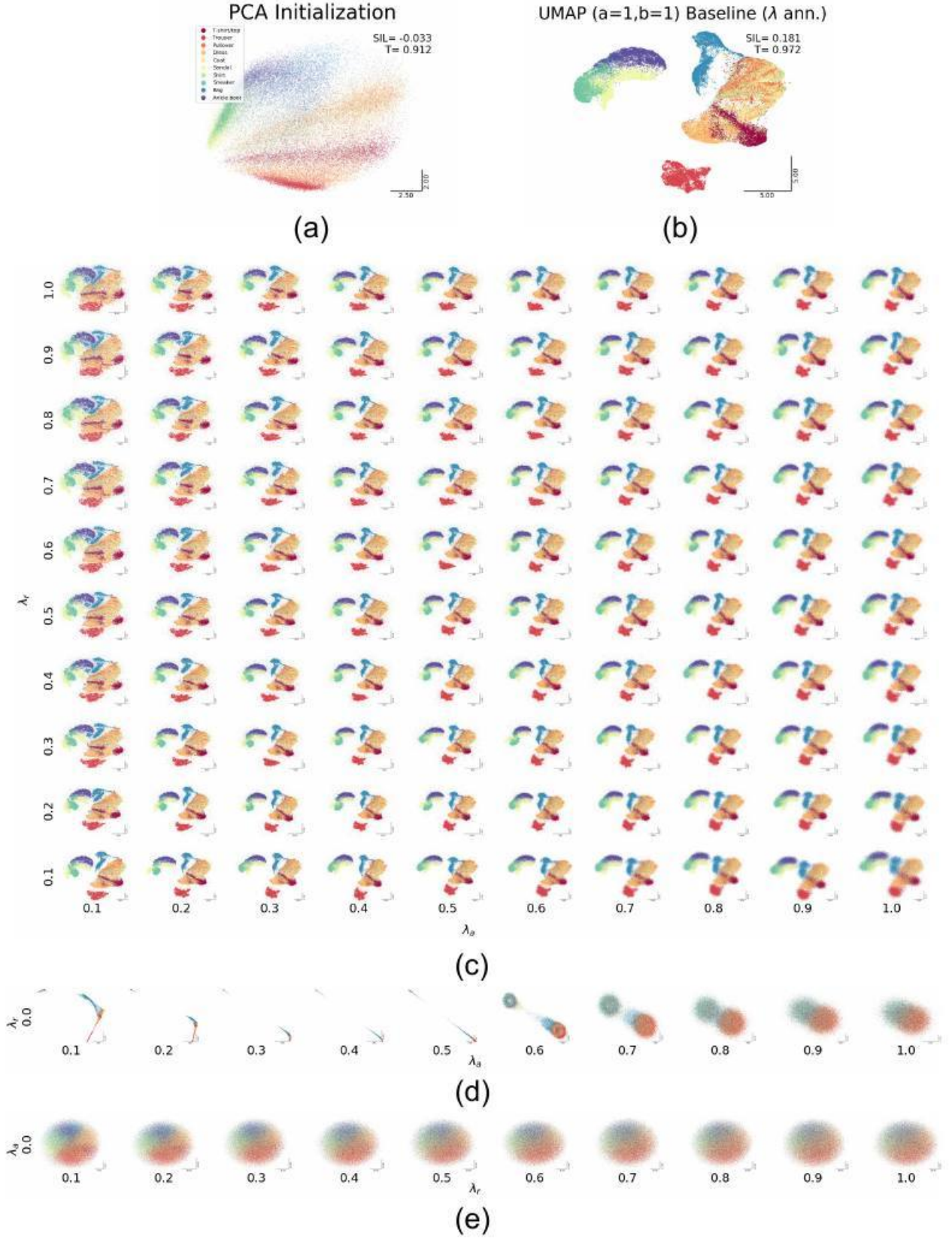


Figure 25: FMNIST detailed shape sweep varying λ_a and λ_r with UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

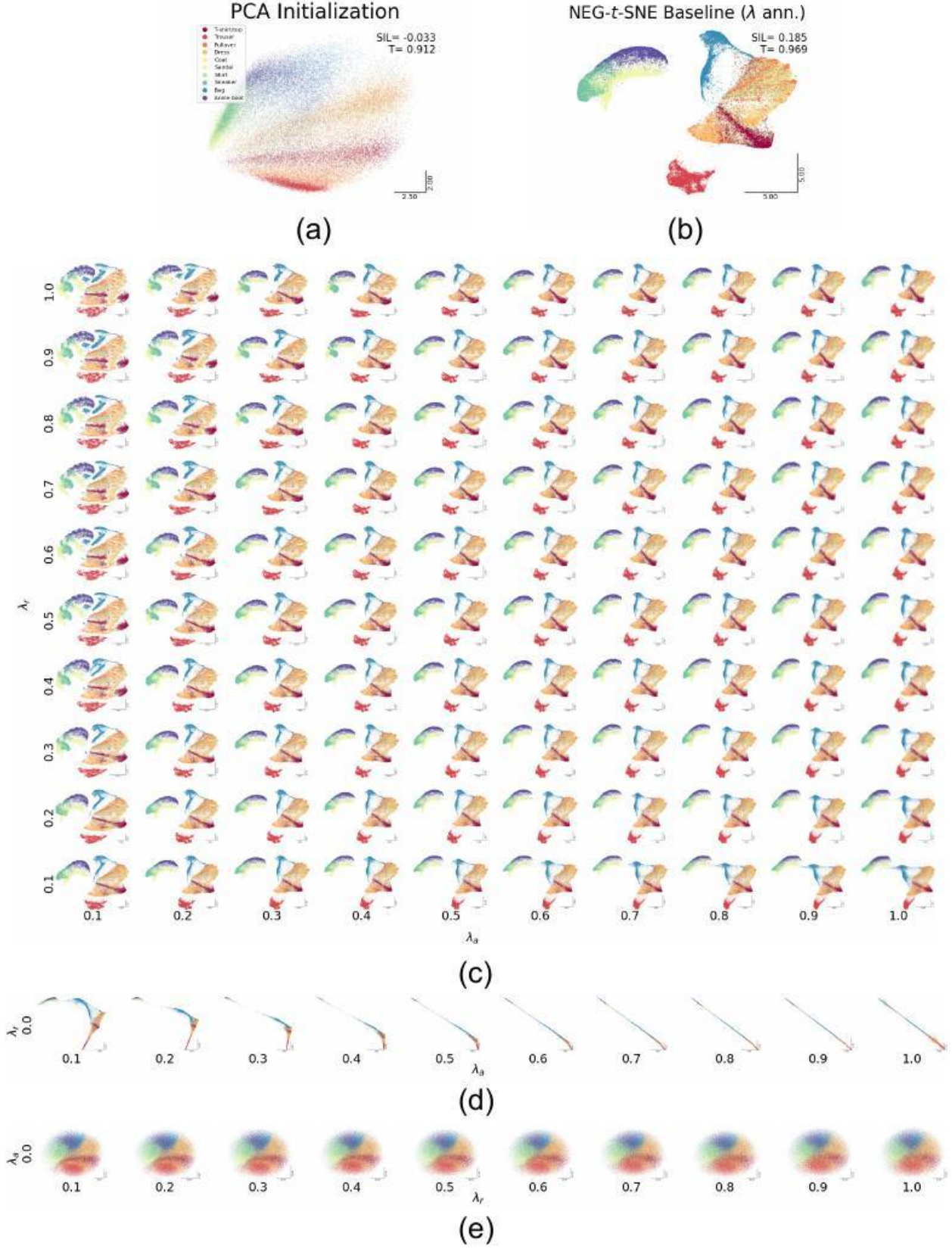


Figure 26: FMNIST detailed shape sweep varying λ_a and λ_r with NEG-t-SNE’s attraction and repulsion shapes. (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

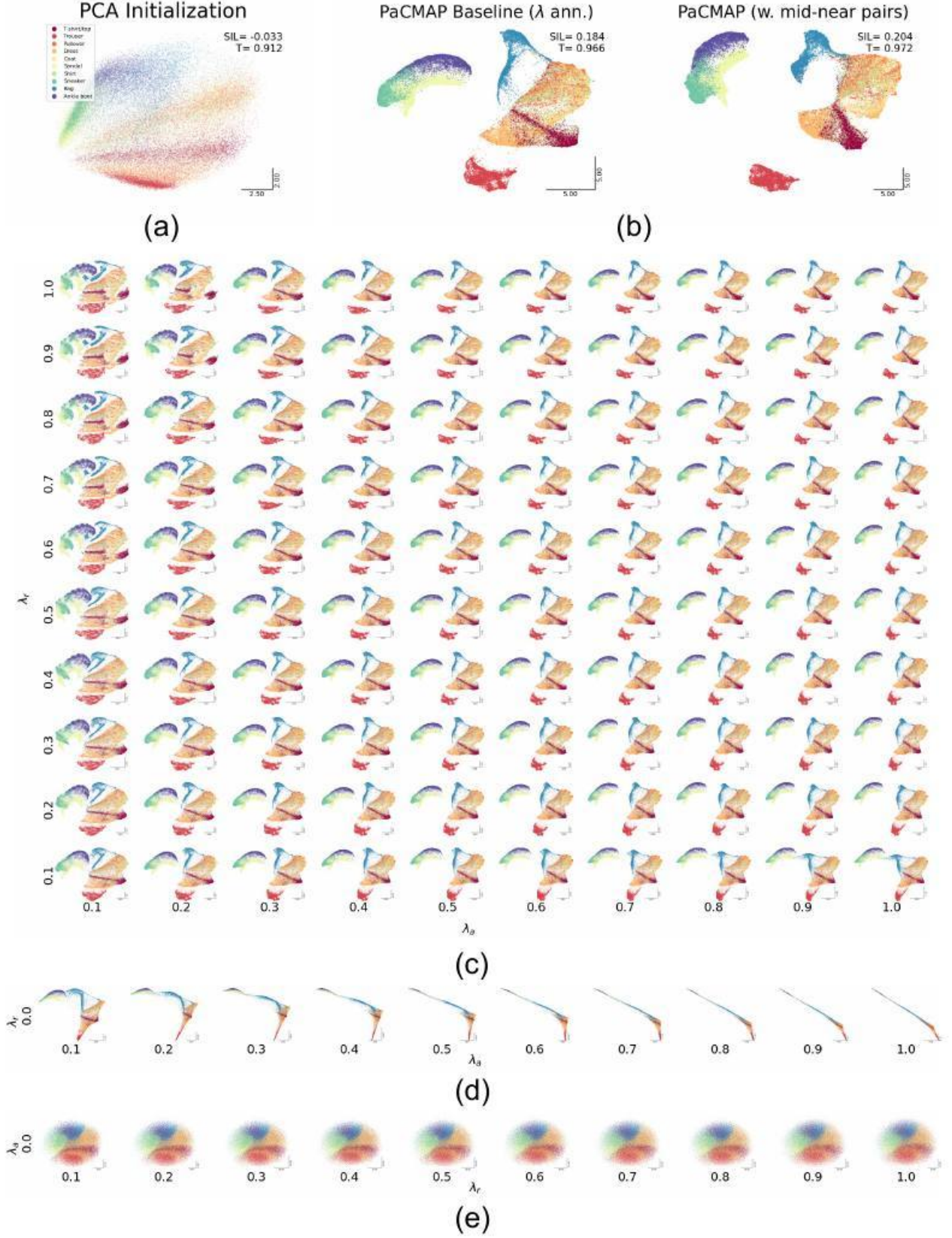


Figure 27: FMNIST detailed shape sweep varying λ_a and λ_r with PaCMAP’s attraction and repulsion shapes. (a) Initialization. (b) Left: annealed- λ baseline without mid-near points to isolate attraction–repulsion; right: baseline with mid-near points included. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

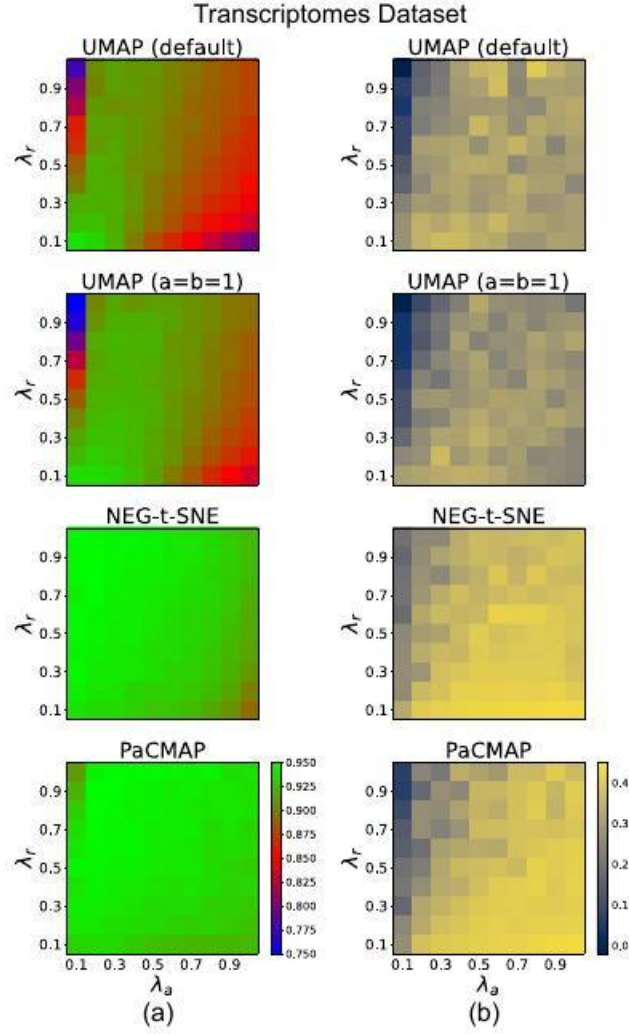


Figure 28: Single-cell transcriptomes shape-sweep summary. (a) Trustworthiness and (b) Silhouette score for different methods under the attraction/repulsion learning-rate sweep.

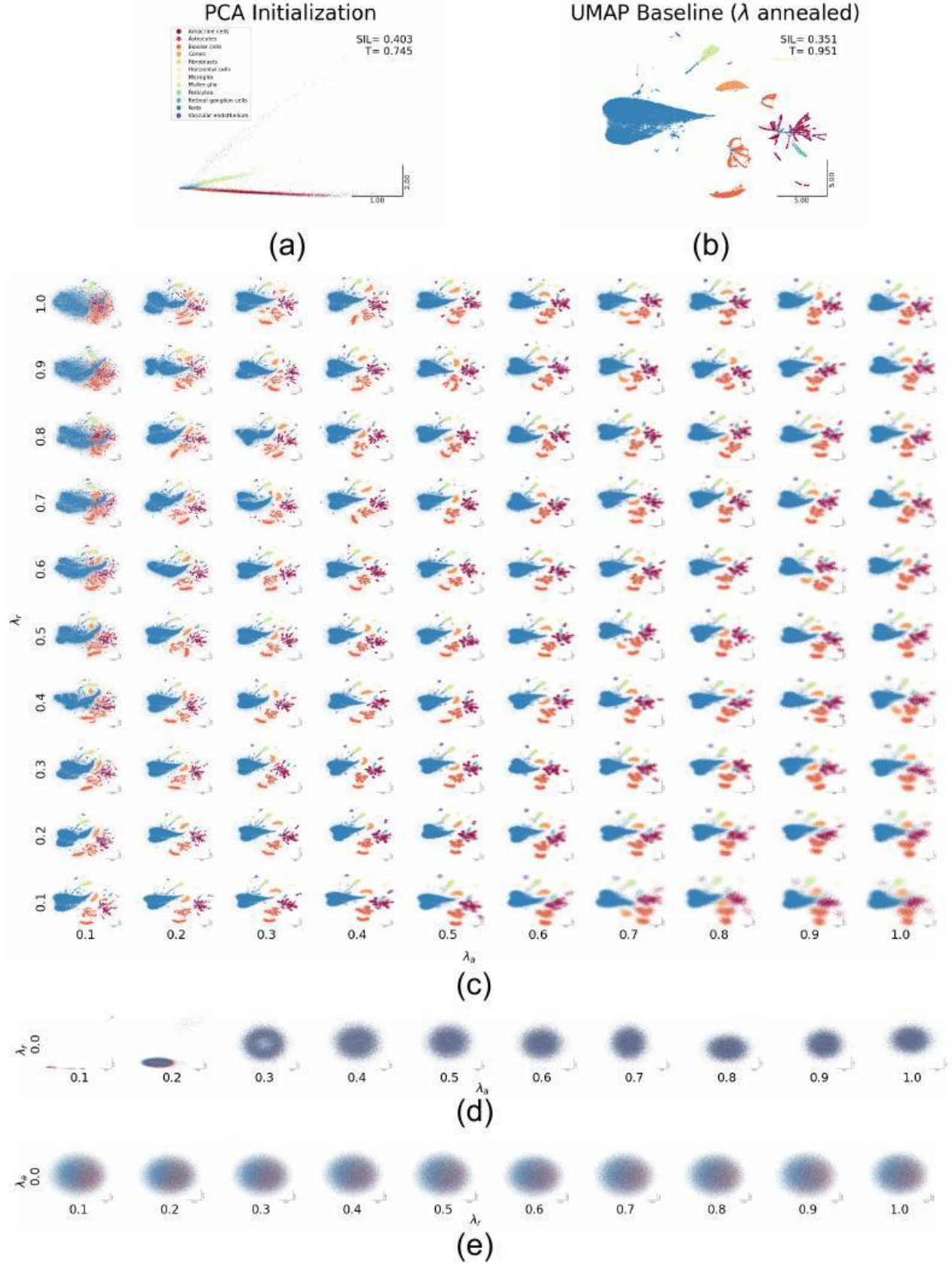


Figure 29: Single-cell transcriptomes detailed shape sweep varying λ_a and λ_r with UMAP's attraction and repulsion shapes. (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

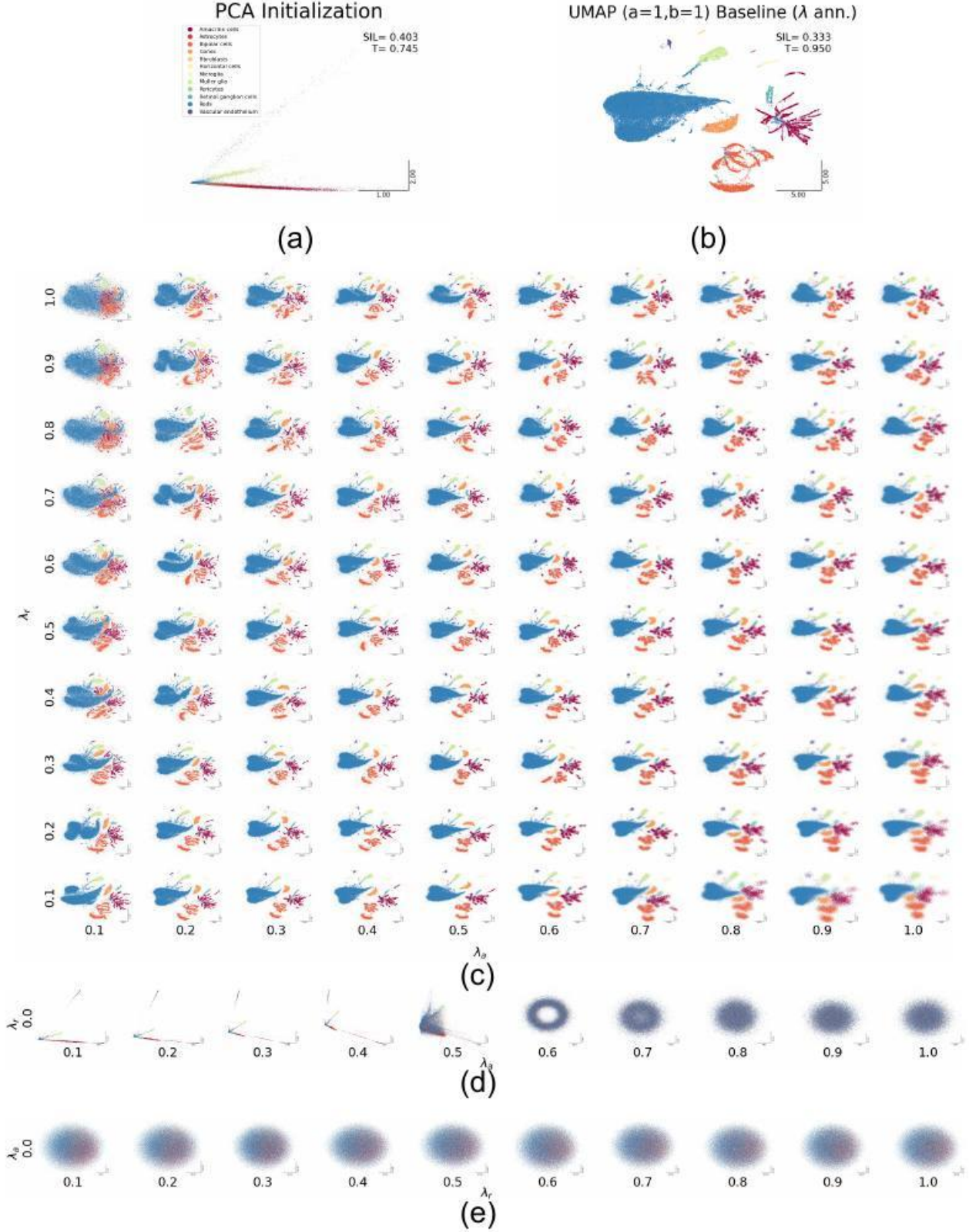


Figure 30: Single-cell transcriptomes detailed shape sweep varying λ_a and λ_r with UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

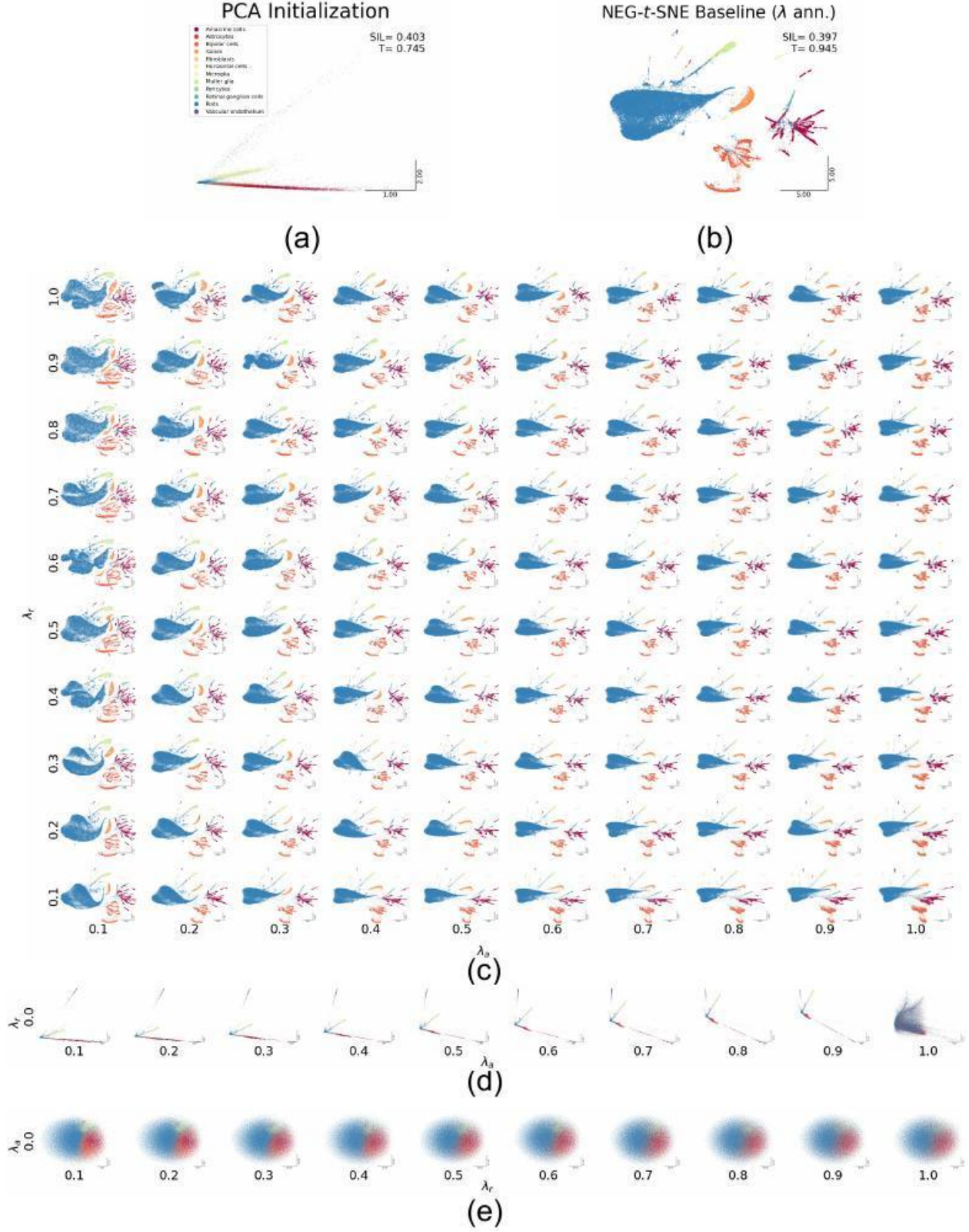


Figure 31: Single-cell transcriptomes detailed shape sweep varying λ_a and λ_r with NEG-t-SNE's attraction and repulsion shapes. (a) Initialization. (b) Annealed- λ baseline. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce distinct clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

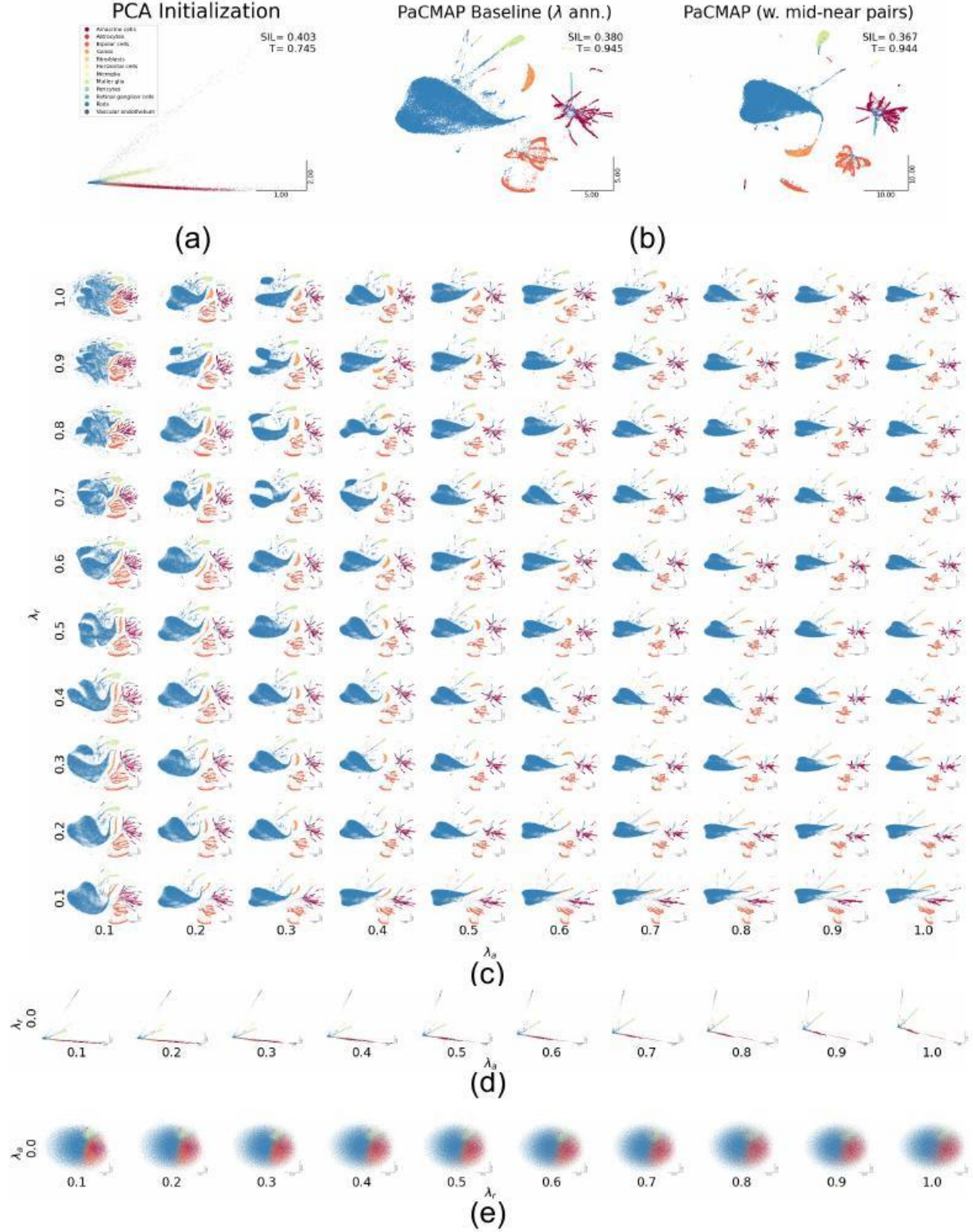


Figure 32: Single-cell transcriptomes detailed shape sweep varying λ_a and λ_r with PaCMAP's attraction and repulsion shapes. (a) Initialization. (b) Left: annealed- λ baseline without mid-near points to isolate attraction–repulsion; right: baseline with mid-near points included. (c) Embeddings from independent λ_a and λ_r variation without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone cannot produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone cannot produce clusters.

J Mix and Match Approach

The shape formalism treats attraction and repulsion as separable design choices, so this appendix tests mix-and-match combinations from different algorithms. Figure 33 pairs attraction and repulsion shapes while keeping the learning rate fixed at $\lambda = 1$ and using the same UMAP optimization, ensuring that differences arise from shape design rather than learning-rate scheduling or optimizer changes. We restrict attraction shapes to $[-1, 0]$ to satisfy Proposition 4.1 and reduce learning-rate sensitivity, and we use finite repulsion shapes to avoid extreme short-distance repulsion. UMAP’s scaled attraction shape ($f_a = 0.5f_a^U$) yields better clustering as measured by silhouette score, whereas PaCMAP’s attraction shape yields better local structure as measured by trustworthiness. The PaCMAP advantage comes from its attraction shape lying within $[-0.5, 0]$, which prevents update flips. Confining UMAP and NEG- t -SNE attraction to $[-0.5, 0]$ ($f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$) improves local structure at the expense of clustering performance (Fig. 34).

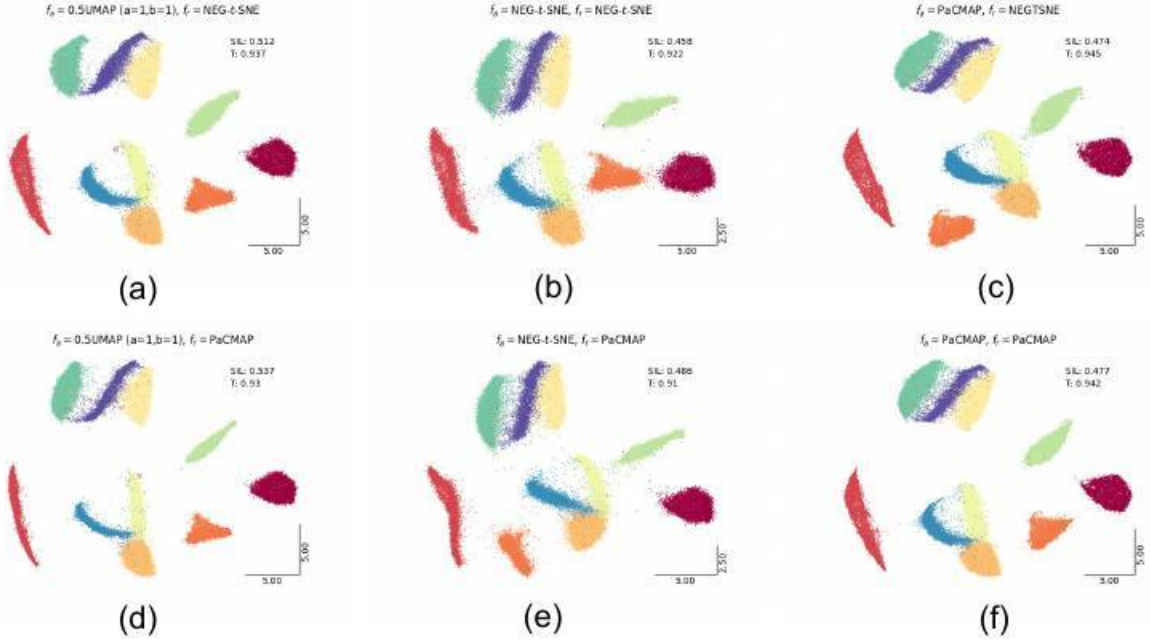


Figure 33: Mix-and-match contribution with constant learning rate ($\lambda = 1$) on MNIST. Attraction shapes satisfy Proposition 4.1 and repulsion shapes are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with NEG- t -SNE repulsion. Bottom row: the same attraction shapes paired with PaCMAP repulsion.

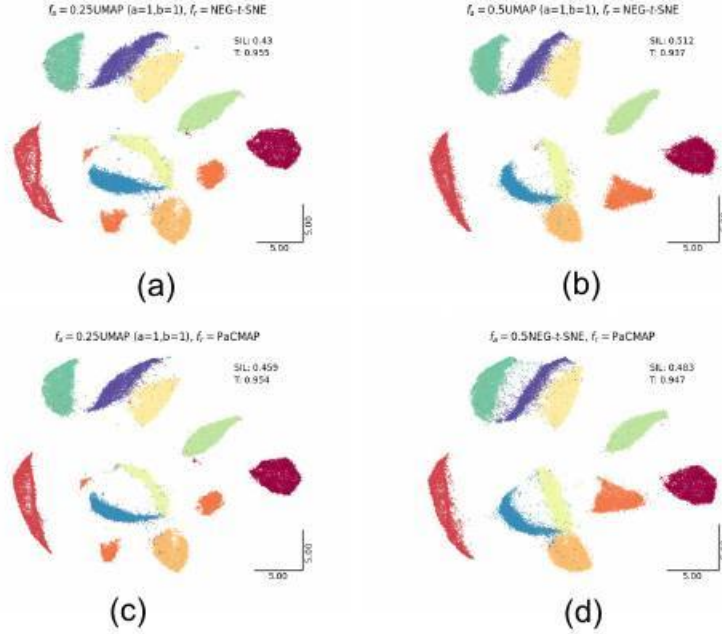


Figure 34: Mix-and-match variants with attraction confined to $[-0.5, 0]$ at constant learning rate ($\lambda = 1$). This prevents flips during attractive updates. Top row: UMAP and NEG- t -SNE attraction shapes with NEG- t -SNE repulsion. Bottom row: UMAP and NEG- t -SNE attraction shapes with PaCMAP repulsion.

K Implementation Details

To make the shape interventions controlled and reproducible, we implemented our own UMAP algorithm. We used `numba` (Lam et al., 2015) to compute an exact, rather than approximate, nearest-neighbor graph with $k = 15$, and we used `scikit-learn`'s (Pedregosa et al., 2011) PCA implementation for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9, and 10. Unless otherwise stated, we used the same implementation when replacing UMAP's attraction and repulsion shapes with those of alternative methods (Figs. 18, 34). Trustworthiness and silhouette scores were computed with the corresponding `scikit-learn` functions.

For visual comparability in the initialization and compactness experiments, the mappings in Figs. 2, 3, 9, 10, and 15 are rotated to the corresponding reference embedding in panel (a). We performed Procrustes alignment by normalizing embeddings to zero mean and unit norm and then using SciPy's (Virtanen et al., 2020) `orthogonal_procrustes` method to obtain rotation and scaling parameters.

For the NEG-*t*-SNE comparison in Fig. 11 we used the original contrastive-embedding framework implementation for both UMAP and Neg-*t*-SNE; the implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22 (b) (right), 27 (b) (right), and 32 (b) (right) use the official PaCMAP package; the implementation details will be released upon acceptance.

The code used to display the PubMed embeddings in Fig. 7 will be released upon acceptance.